

DEEP LEARNING APPROACH FOR KIDNEY CT SEGMENTATION USING ATTENTION U-NET++

A

Project Report

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Requirements for the award of the Degree of

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DECLARATION BY CANDIDATE

We, **V.VIGNESH** bearing hall ticket number **1602-22-737-060**, and **T.LASYA PRIYA** bearing hall ticket number **1602-22-737-301**, hereby declare that the project report entitled “**Deep Learning Approach for Kidney CT Segmentation Using Attention U-Net++**” under the guidance of **Mrs. DRL Prasanna**, Assistant Professor, Department of Information Technology, Vasavi College of Engineering, Hyderabad, is submitted in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering in Information Technology**.

This is a record of bonafide work carried out by us and the results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

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BONAFIDE CERTIFICATE

This is to certify that the project entitled “**Deep Learning Approach for Kidney CT Segmentation Using Attention U-Net++**” being submitted by **V.VIGNESH** bearing hall ticket number **1602-22-737-060** and **T.LASYA PRIYA** bearing hall ticket number **1602-22-737-301** in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Information Technology, is a record of bonafide work carried out by them under my guidance.

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ABSTRACT

Kidney segmentation from medical images plays a crucial role in diagnosis, treatment planning, and disease monitoring. Accurate identification of kidney regions is essential for detecting abnormalities such as tumors, cysts, and other renal diseases. However, manual segmentation performed by radiologists is time-consuming, labor-intensive, and subject to inter-observer variability. This highlights the need for efficient and automated segmentation techniques in modern clinical workflows.

This project presents a deep learning-based approach for kidney segmentation using an Attention U-Net++ architecture. The proposed model combines the strengths of U-Net++, attention mechanisms, and deep supervision to enhance segmentation performance. The nested skip connections in U-Net++ reduce the semantic gap between encoder and decoder features, while attention gates enable the model to focus on relevant kidney regions and suppress irrelevant background information.

The system is trained and evaluated on benchmark datasets such as CHAOS and KiTS19, which consist of annotated CT and MRI images. Various preprocessing techniques, including normalization and resizing, are applied to ensure data consistency. Additionally, data augmentation methods such as rotation, flipping, and scaling are used to increase dataset diversity and improve model generalization.

To optimize performance, the model is trained using a combination of Binary Cross-Entropy (BCE) loss and Dice loss, which effectively handle class imbalance and improve overlap accuracy. Validation strategies such as patient-wise data splitting and cross-validation are employed to ensure robustness and avoid overfitting.

Experimental results demonstrate that the proposed model achieves high segmentation accuracy, with a Dice Similarity Coefficient (DSC) of 0.975 on the CHAOS dataset. The model shows consistent performance across different CT slices, including challenging cases with low contrast and partial organ visibility.

Overall, the proposed system provides an efficient and reliable solution for automated kidney segmentation. It can serve as a valuable tool in clinical applications, supporting medical professionals in diagnosis, treatment planning, and further analysis such as tumor detection and disease classification in medical imaging.

TABLE OF CONTENTS

DECLARATION BY THE CANDIDATE	i
BONAFIDE CERTIFICATE	ii
ACKNOWLEDGEMENT	iii
ABSTRACT	iv
TABLE OF CONTENTS	v
LIST OF FIGURES	vii
LIST OF TABLES	viii
LIST OF ABBREVIATIONS	ix
1 INTRODUCTION	1
1.1 Background	1
1.2 Problem Statement	2
1.3 Objectives	2
1.4 Scope of the Project	3
1.5 Methodology Overview	3
1.6 Report Organization	4
2 LITERATURE SURVEY	5
2.1 Introduction	6
2.2 Existing Systems / Related Work	6
2.3 Limitations of Existing Systems	6
2.4 Motivation for Proposed System	6
2.5 Summary	6
3 IMPLEMENTATION	25
3.1 Introduction	25
3.2 Tools and Platforms	25
3.3 Proposed Methodology	25
3.4 Architecture Diagram	27
3.5 Algorithm / Mathematical Modelling	31
3.6 System Workflow / Process Flow	32
4 EXPERIMENTATION	34
4.1 System Configuration	34
4.2 Dataset Description	35
4.3 Evaluation Metrics	36
5 RESULTS AND DISCUSSION	37

5.1	Result Analysis	37
5.2	Output Screenshots	42
6	CONCLUSION AND FUTURE SCOPE	46
6.1	Conclusion	46
6.2	Future Scope	47
7	REFERENCES	48
8	APPENDIX	50
9	PUBLICATION PAPER DRAFT	58
10	PROGRAM OUTCOMES (PO)	60
11	PROGRAM SPECIFIC OUTCOMES (PSO)	61
12	SUSTAINABLE DEVELOPMENT GOALS (SDG)	62

LIST OF FIGURES

3.1	Overall Architecture of the Proposed Attention U-Net++ with Deep Supervision	27
3.2	Attention Gate Mechanism Used in Skip Connections	29
3.3	Deep Supervision Applied at Multiple Decoder Stages	30
3.4	System Workflow of Kidney Segmentation	33
4.1	Sample CT Image and Corresponding Ground Truth Kidney Mask .	35
5.1	Comparison of Dice Scores Across Different Models	39
5.2	Training Loss Convergence	40
5.3	Dice Score Progression During Training	41
5.4	Input Image, Ground Truth Mask, and Predicted Segmentation . . .	42
5.5	Visualization of Kidney Segmentation Results Showing Original CT Scans, Doctor-Annotated Ground Truth, and AI Predicted Masks . .	43
5.6	Segmentation Results on Multiple CT Slices Showing Comparison Between Input Images, Ground Truth, and Predicted Outputs	44
5.7	Qualitative Comparison of CT Images, Ground Truth Masks, and Predicted Segmentation Outputs	45
8.1	Similarity Report Overview	56
8.2	Similarity Report Overview	57
9.1	First Page of the Publication Paper Draft – Attention U-Net++ for Kidney CT Segmentation	59

LIST OF TABLES

3.1	Architecture Details of Proposed Model	32
4.1	Training Parameters	36
5.1	Comparison of Dice Scores Across Models	38
5.2	Performance Metrics Comparison on CHAOS CT Dataset	38
5.3	Ablation Study on CHAOS CT Dataset	38
10.1	PO Mapping Table	60
11.1	PSO Mapping Table	61
12.1	SDG Goals Mapping Table	62

LIST OF ABBREVIATIONS

CNN	Convolutional Neural Network
CT	Computed Tomography
DSC	Dice Similarity Coefficient
IoU	Intersection over Union
U-Net	Convolutional Network for Biomedical Segmentation
U-Net++	Nested U-Net Architecture
ReLU	Rectified Linear Unit
BCE	Binary Cross Entropy
GPU	Graphics Processing Unit
ML	Machine Learning
DL	Deep Learning
CHAOS	Combined Healthy Abdominal Organ Segmentation

1. INTRODUCTION

1.1 Background

Medical imaging plays a vital role in modern healthcare for diagnosis, treatment planning, and disease monitoring. Among various organs, the kidney is particularly important as it performs essential functions such as filtration of blood, removal of waste products, and regulation of body fluids. Accurate analysis of kidney structures in medical images such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) is crucial for detecting abnormalities, including tumors, cysts, and other renal diseases. Early and precise identification of such conditions can significantly improve patient outcomes and support effective clinical decision-making.

Traditionally, kidney segmentation is performed manually by radiologists. However, manual segmentation is time-consuming, subjective, and prone to human error, especially when dealing with large volumes of medical data. Additionally, variations in image quality, noise, intensity inhomogeneity, and anatomical differences among patients make the segmentation process more complex and less reliable. These limitations highlight the need for automated and efficient segmentation techniques.

With the rapid advancement of Artificial Intelligence and deep learning, automated medical image segmentation has gained significant attention. Convolutional Neural Networks (CNNs) have demonstrated exceptional capability in learning hierarchical and spatial features from medical images. Architectures such as U-Net [1] and its advanced variants, including U-Net++ [2] and Attention U-Net [3], have become widely adopted for biomedical image segmentation due to their ability to combine low-level and high-level features effectively.

This project focuses on developing an automated kidney segmentation system using an advanced deep learning architecture called Attention U-Net++. The proposed model integrates attention mechanisms, nested skip connections, and deep supervision to improve segmentation accuracy and robustness. It aims to accurately identify kidney regions in CT and MRI images while effectively handling challenges such as low contrast, complex boundaries, noise, and background interference.

Furthermore, the proposed approach is designed to provide consistent and reliable segmentation results across different imaging conditions and patient variations.

1.2 Problem Statement

Kidney segmentation from medical images is a challenging task due to several factors:

- **Low contrast:** Kidney regions often have similar intensity values as surrounding tissues, making it difficult to distinguish boundaries.
- **Anatomical variability:** The shape, size, and position of kidneys vary across different patients.
- **Complex boundaries:** Irregular shapes and unclear edges make segmentation difficult.
- **Background noise:** Presence of other organs and structures interferes with accurate detection.

Existing methods either rely heavily on manual intervention or fail to achieve high accuracy and generalization. Therefore, there is a need for an efficient and automated system that can accurately segment kidney regions while minimizing errors.

1.3 Objectives

The main objectives of this project are:

- To develop a deep learning-based kidney segmentation model using Attention U-Net++ architecture.
- To improve segmentation accuracy by incorporating attention mechanisms and nested skip connections.
- To handle challenges such as low contrast, anatomical variability, and background noise.
- To apply data preprocessing and augmentation techniques to enhance model performance.
- To evaluate the model using appropriate performance metrics such as Dice Similarity Coefficient (DSC).
- To provide a foundation for future extension towards tumor detection and disease analysis.

1.4 Scope of the Project

The scope of this project includes the development of an automated system for kidney segmentation from medical images. The system focuses on:

- Processing CT and MRI images from the CHAOS dataset.
- Segmenting kidney regions using deep learning techniques.
- Evaluating the performance of the model using standard metrics.

However, the current implementation is limited to kidney segmentation only. Detection of tumors and other abnormalities is considered as future work and is not fully implemented in the present system.

1.5 Methodology Overview

The proposed system follows a deep learning-based approach for kidney segmentation. The overall methodology includes the following steps:

1. **Data Collection:** Medical images are obtained from the CHAOS dataset.
2. **Preprocessing:** Images are normalized, resized, and augmented to improve training efficiency.
3. **Model Design:** An Attention U-Net++ architecture is implemented for segmentation.
4. **Training:** The model is trained using labeled images and corresponding segmentation masks.
5. **Validation:** Techniques such as cross-validation and patient-wise splitting are used.
6. **Evaluation:** Performance is measured using metrics such as Dice Similarity Coefficient.

The integration of attention mechanisms helps the model focus on relevant regions, while nested skip connections improve feature propagation and segmentation accuracy.

1.6 Report Organization

This report is organized into multiple chapters for the systematic and structured presentation of the work carried out in this project:

- **Chapter 1: Introduction** – Presents the background of medical image segmentation, highlights the importance of kidney segmentation, defines the problem statement, and outlines the objectives and scope of the project.
- **Chapter 2: Literature Survey** – Reviews existing research work related to kidney segmentation using deep learning techniques and identifies the limitations and research gaps in current approaches.
- **Chapter 3: Implementation** – Describes the proposed methodology in detail, including the Attention U-Net++ architecture, attention mechanisms, deep supervision, and the tools and technologies used for implementation.
- **Chapter 4: Experimentation** – Provides details of the dataset used, pre-processing steps, training configuration, evaluation metrics, and experimental setup.
- **Chapter 5: Results and Discussion** – Presents the qualitative and quantitative results obtained from the model, along with detailed analysis and comparison with existing methods.
- **Chapter 6: Conclusion and Future Scope** – Summarizes the overall work, highlights key findings, and suggests possible improvements and future research directions.

2. LITERATURE SURVEY

Paper 1: Comparative Study of Deep Learning Semantic Segmentation Models for Kidney Segmentation

Introduction:

Kidney segmentation is important for detecting renal diseases and supporting treatment planning. Ultrasound imaging is widely used due to its low cost, but it suffers from noise, poor contrast, and variability, making manual segmentation difficult.

Existing Systems / Related Work:

Daoud et al. (2025) compared eleven deep learning models including U-Net [1], UNet++ [2], DeepLabV3+, and SegFormer. The models were trained using the Open Kidney Ultrasound Dataset with cross-validation and evaluated using accuracy, precision, recall, F1-score, and IoU. Transformer-based models achieved higher accuracy, while attention-based models improved boundary detection[3].

Limitations of Existing Systems:

- Does not consider CT-based segmentation
- Lacks integration of attention, deep supervision, and nested architecture
- Limited anatomical detail compared to CT images

Motivation for Proposed System:

These limitations highlight the need for models that can effectively handle CT images with greater precision and detail. Integrating attention mechanisms, deep supervision, and nested skip connections can significantly improve feature extraction, boundary localization, and overall segmentation accuracy. Such enhancements enable the model to better capture fine structural details and handle complex variations in medical images.

Summary:

The study highlights the effectiveness of existing segmentation models; however, it lacks comprehensive CT-based analysis and advanced feature integration. This creates the need for improved architectures such as Attention U-Net++, which provide better accuracy, robustness, and suitability for medical image segmentation tasks.

Paper 2: U-Net Deep Learning Architecture for Segmentation of Vascular and Non-Vascular Images

2.1 Introduction

Medical image segmentation is important for identifying anatomical structures and supporting diagnosis and treatment planning. Accurate segmentation helps in detecting abnormalities and improving clinical decision-making. U-Net is widely used due to its encoder–decoder structure[1] and ability to capture both spatial and contextual information.

2.2 Existing Systems / Related Work

Suri et al. (2023) conducted a systematic review of 267 U-Net-based studies. The architectures were categorized into conventional, attention-based, residual, hybrid, and ensemble models. The study analyzed components such as encoder design, skip connections, loss functions, and optimization techniques. Attention-based and hybrid models showed improved segmentation accuracy[3] and better feature extraction. The review also highlighted the importance of data preprocessing and augmentation in enhancing model performance.

2.3 Limitations of Existing Systems

- No specific model for kidney segmentation
- Lack of CT-based experimental validation
- Does not focus on organ-specific challenges
- Limited integration of deep supervision techniques

2.4 Motivation for Proposed System

These limitations indicate the need for a task-specific model that can effectively handle kidney segmentation. Combining attention mechanisms, deep supervision, and nested skip connections can improve accuracy and robustness, especially for CT-based images.

2.5 Summary

The study provides a comprehensive overview of U-Net variants and their improvements. However, the lack of CT-based implementation and integration of advanced techniques motivates the development of improved models like Attention U-Net++.

Paper 3: MLP-UNet for Glomerulus Segmentation

Introduction:

Glomerulus segmentation plays an important role in diagnosing kidney diseases using histopathology images. Accurate identification helps in analyzing renal structures and detecting abnormalities. However, manual segmentation is time-consuming and prone to variability among experts, making automated deep learning methods essential.

Existing Systems / Related Work:

Saikia et al. (2023) proposed MLP-UNet, which replaces convolutional encoders with MLP-based encoders such as MLP-Mixer and ResMLP. The model follows an encoder–decoder architecture with skip connections, where a CNN-based embedding layer extracts initial features and MLP modules capture global contextual information. It was trained using Dice loss and evaluated on the HuBMAP dataset using Dice coefficient and IoU, achieving comparable performance with fewer parameters.

Limitations of Existing Systems:

- Limited to 2D histopathology images
- Does not incorporate attention mechanisms[3]
- Lacks deep supervision for better feature learning
- Not applicable for CT-based kidney segmentation

Motivation for Proposed System:

These limitations highlight the need for a more advanced model capable of effectively handling CT images with low contrast, noise, and anatomical variability. Integrating attention mechanisms, deep supervision, and nested skip connections can significantly improve feature representation, boundary localization, and overall segmentation accuracy. Such a combination enables the model to focus on relevant regions while learning multi-scale contextual information, making it more suitable for medical image segmentation tasks.

Summary:

MLP-UNet improves computational efficiency and reduces model complexity; however, it lacks advanced mechanisms required for precise kidney segmentation. In particular, the absence of attention and deep feature integration limits its ability to accurately capture complex anatomical structures. This motivates the use of improved architectures such as Attention U-Net++, which provide better accuracy, robustness, and generalization for kidney segmentation in CT images.

Paper 4: Feature Aggregation and Edge-Enhanced Attention Model

Introduction:

Accurate image segmentation is challenging due to noise, low contrast, and unclear boundaries. In medical imaging, precise boundary detection is essential for identifying affected regions and supporting diagnosis. Traditional models often fail to capture fine details, especially in complex and noisy images.

Existing Systems / Related Work:

Wang et al. (2023) proposed a segmentation model combining CNN-based ResNeXt and transformer-based Swin Transformer to capture both local and global features. A Network Feature Aggregation (NFA) module is used to fuse multi-level features, while an Edge-Enhanced Attention mechanism improves boundary detection[3]. Additional modules such as CBAM and CASPP enhance multi-scale feature learning. The model was evaluated on the ISIC2018 dataset and showed improved segmentation performance compared to baseline models.

Limitations of Existing Systems:

- Applied only to skin lesion images
- Not evaluated on CT or MRI datasets
- Does not include deep supervision
- Lacks nested skip connections

Motivation for Proposed System:

This work highlights the importance of attention mechanisms and feature aggregation in improving segmentation accuracy [9][10]. However, kidney segmentation from CT images involves additional challenges such as low contrast, anatomical variability, and complex boundaries, which require more specialized architectures. Therefore, a model that combines attention mechanisms, nested skip connections, and deep supervision is essential to achieve accurate and robust segmentation. These components enable better feature representation, improved localization, and enhanced learning across multiple scales.

Summary:

The reviewed model improves segmentation performance through advanced feature aggregation and attention mechanisms. However, its limitation in handling complex medical imaging scenarios and organ-specific variations creates the need for more robust approaches. This motivates the development of improved kidney segmentation models such as Attention U-Net++, which offer better accuracy, generalization, and reliability for medical image analysis.

Paper 5: Attention-Based Residual U-Net for Segmentation

Introduction:

Accurate segmentation in complex environments requires models that can effectively capture important features and maintain stable training. Traditional deep learning models often face issues such as vanishing gradients and limited feature representation, especially in challenging conditions.

Existing Systems / Related Work:

Jing et al. (2023) proposed an Attention-Based Residual U-Net that integrates Convolutional Block Attention Modules (CBAM) and residual connections into the U-Net architecture[10]. Residual blocks help in improving gradient flow and preventing degradation, while attention modules focus on relevant regions by suppressing background noise. The model was evaluated on road crack detection datasets and showed improved segmentation accuracy and better feature representation compared to standard U-Net.

Limitations of Existing Systems:

- Applied to non-medical datasets
- Not validated on CT or MRI images
- Does not include deep supervision
- Lacks nested skip connections

Motivation for Proposed System:

This work highlights the importance of attention mechanisms and residual learning in improving segmentation performance [3]. However, integrating these techniques with nested architectures and deep supervision can further enhance accuracy, robustness, and feature representation in medical image segmentation tasks. Such combinations enable better localization of target regions, improved boundary delineation, and more stable training across complex datasets.

Summary:

The model demonstrates improved feature extraction and segmentation performance; however, its limitations in handling complex medical imaging scenarios and lack of advanced feature integration highlight the need for improved architectures such as Attention U-Net++. These advanced models provide better localization, enhanced feature fusion, and improved segmentation accuracy, making them more suitable for medical image analysis.

Paper 6: CBAM-DC-UNet for White Blood Cell Segmentation

Introduction:

Segmentation of white blood cells is important for diagnosing diseases such as leukemia and other blood-related disorders. However, medical images often contain noise, overlapping cells, and unclear boundaries, making accurate segmentation challenging[1]. Traditional methods fail to generalize well under such conditions.

Existing Systems / Related Work:

Li et al. (2023) proposed CBAM-DC-UNet, an improved U-Net architecture that integrates dilated convolutions and Convolutional Block Attention Modules (CBAM)[3]. Dilated convolutions expand the receptive field without reducing resolution, while attention modules enhance feature selection by focusing on important regions. The model was trained using the RAdam optimizer and evaluated using IoU, recall, and accuracy, showing improved performance over standard U-Net.

Limitations of Existing Systems:

- Limited to microscopic image datasets
- Not applicable to CT-based organ segmentation
- Does not include deep supervision
- Lacks nested skip connections

Motivation for Proposed System:

Although this model improves segmentation performance through attention mechanisms and dilated convolutions, it is not well-suited for complex organ-level segmentation tasks such as kidneys. Kidney segmentation from CT images requires the ability to capture multi-scale features, handle anatomical variability, and accurately delineate complex boundaries. Therefore, there is a need for models that can effectively integrate multi-scale feature extraction with robust contextual understanding. Combining attention mechanisms with nested skip connections and deep supervision can significantly enhance feature learning, improve boundary detection, and provide better generalization for kidney segmentation tasks.

Summary:

CBAM-DC-UNet improves segmentation accuracy by combining attention mechanisms and dilated convolutions. However, its focus on microscopic image datasets and lack of advanced architectural integration limit its applicability to kidney segmentation in CT images. These limitations highlight the need for more robust and scalable models such as Attention U-Net++, which are better suited to handle complex medical imaging scenarios with higher accuracy and reliability.

Paper 7: MAGRes-UNet for Medical Image Segmentation

Introduction:

Medical image segmentation is essential for accurate diagnosis and treatment planning. However, complex anatomical structures and variations in medical images make segmentation challenging. Traditional models often struggle with issues such as vanishing gradients and inefficient feature fusion[10].

Existing Systems / Related Work:

Hussain and Shouno (2024) proposed MAGRes-UNet, which integrates multi-attention gating and residual learning into the U-Net architecture. Multi-attention gates help the model focus on relevant spatial and contextual features, while residual blocks improve gradient flow and training stability[3]. The model was evaluated on medical imaging datasets such as MRI and skin lesion datasets, showing improved segmentation accuracy and robustness compared to baseline models.

Limitations of Existing Systems:

- Not evaluated on CT-based kidney datasets
- Does not include deep supervision
- Lacks nested skip connections
- Limited generalization for kidney segmentation tasks

Motivation for Proposed System:

This work demonstrates that attention mechanisms and residual learning can significantly improve segmentation performance by enhancing feature extraction and stabilizing training. However, for kidney segmentation, it is equally important to incorporate multi-scale feature fusion and efficient gradient propagation to handle complex anatomical structures and variations in CT images. Integrating attention mechanisms with nested skip connections and deep supervision can further enhance segmentation accuracy by enabling better feature representation, improved boundary localization, and robust learning across multiple scales.

Summary:

MAGRes-UNet improves segmentation performance by combining attention mechanisms with residual learning, resulting in better feature extraction and training stability. However, the absence of deep supervision and nested architecture limits its ability to fully capture multi-scale contextual information and refine segmentation outputs. This motivates the development of more advanced models such as Attention U-Net++, which integrate multiple techniques to achieve improved accuracy, robustness, and reliability for kidney segmentation tasks.

Paper 8: EfficientNet-Based U-Net for Kidney Tumor Segmentation

Introduction:

Kidney tumor segmentation is crucial for early diagnosis and treatment planning in renal diseases. CT imaging is widely used as it provides detailed anatomical information[1]. However, manual segmentation is time-consuming and prone to variability, making automated deep learning approaches essential.

Existing Systems / Related Work:

Abdelrahman and Viriri (2023) proposed EfficientNet-based U-Net models for kidney tumor segmentation. EfficientNet architectures (B0–B7) were used as encoder backbones to improve feature extraction, while the decoder followed the standard U-Net structure. The model was trained and evaluated on the KiTS19 dataset using Intersection over Union (IoU) as the primary metric. The results showed that EfficientNet encoders effectively capture both global and local features, achieving high segmentation accuracy for kidney and tumor regions.

Limitations of Existing Systems:

- Does not include attention mechanisms
- Lacks nested skip connections
- Does not incorporate deep supervision

Motivation for Proposed System:

Although EfficientNet improves feature extraction through efficient scaling and deeper representations, segmentation performance can be further enhanced by integrating advanced architectural techniques. Attention mechanisms enable the model to focus on relevant kidney regions while suppressing background noise. Additionally, nested skip connections and deep supervision improve multi-scale feature learning and facilitate better gradient propagation. These enhancements are particularly important for accurately capturing complex kidney structures and tumor boundaries in CT images.

Summary:

EfficientNet-based U-Net models achieve high segmentation accuracy due to their strong feature extraction capability. However, the absence of attention mechanisms and advanced architectural integration limits their ability to effectively handle complex medical imaging scenarios. This motivates the development of more comprehensive models such as Attention U-Net++, which combine multiple techniques to achieve improved accuracy, robustness, and reliability for kidney segmentation tasks.

Paper 9: Attention-Powered U-Net for Kidney Tumor Segmentation

Introduction:

Kidney tumor segmentation is essential for accurate diagnosis and effective treatment planning. CT imaging provides detailed anatomical information, but segmenting tumor regions is challenging due to low contrast and irregular tumor shapes. Manual segmentation is time-consuming and prone to inconsistencies, highlighting the need for automated solutions.

Existing Systems / Related Work:

Elhadi and Bayoub (2025) proposed an attention-powered U-Net architecture to improve kidney tumor segmentation. The model integrates attention gates into the encoder–decoder structure, allowing it to focus on tumor-relevant regions while suppressing background features[3]. It was trained on the KiTS19 dataset and evaluated using Dice coefficient and IoU. The approach improved tumor localization and reduced false positives compared to standard U-Net.

Limitations of Existing Systems:

- Does not include nested skip connections
- Lacks deep supervision
- Limited exploration of multi-scale feature learning
- Performance depends on attention module design

Motivation for Proposed System:

Although attention mechanisms improve the model’s ability to focus on relevant regions, they alone are not sufficient to handle the complexity of kidney structures and tumor boundaries. Accurate segmentation of CT images requires effective multi-scale feature fusion and robust contextual understanding. Integrating attention mechanisms with nested skip connections and deep supervision can significantly enhance feature representation, improve boundary delineation, and increase overall segmentation accuracy [1]. This combination is particularly important for detecting tumors with irregular shapes and varying intensities in medical images.

Summary:

The attention-powered U-Net improves tumor segmentation by emphasizing important regions and reducing background noise. However, the lack of advanced architectural integration limits its ability to capture complex anatomical variations and multi-scale features. This motivates the development of more robust models such as Attention U-Net++, which combine multiple techniques to achieve improved accuracy, stability, and reliability for kidney and tumor segmentation tasks.

Paper 10: Ensemble U-Net Models for Kidney Tumor Segmentation

Introduction:

Kidney tumor segmentation is important for cancer diagnosis, treatment planning, and prognosis. CT imaging provides detailed visualization of kidney structures, but accurate segmentation is challenging due to variations in tumor shape and intensity. Manual segmentation is time-consuming and prone to inconsistencies.

Existing Systems / Related Work:

Causey et al. (2022) proposed an ensemble approach using multiple U-Net models for kidney tumor segmentation. Two U-Net models were trained separately—one for kidney and tumor masks, and another for combined segmentation[1]. The outputs were combined using a voting mechanism to improve robustness. The model was trained on the KiTS19 dataset with preprocessing techniques such as normalization and data augmentation. Post-processing methods were applied to refine segmentation results.

Limitations of Existing Systems:

- Based on conventional U-Net architecture
- Uses 2D slice-based segmentation
- Does not include attention mechanisms
- Lacks deep supervision and nested connections

Motivation for Proposed System:

While ensemble methods improve robustness and segmentation accuracy by combining multiple models, they significantly increase computational complexity and do not directly address architectural limitations within a single model. For practical clinical applications, there is a need for a unified and efficient architecture that can achieve high accuracy without relying on multiple models. Integrating attention mechanisms, deep supervision, and nested skip connections within a single framework can enhance feature learning, improve boundary localization, and maintain computational efficiency. Such an approach provides a balanced solution for achieving both accuracy and scalability in kidney segmentation tasks.

Summary:

The ensemble U-Net approach improves segmentation accuracy through model combination and post-processing techniques. However, its dependence on multiple models and lack of advanced architectural integration reduce its efficiency and scalability. This motivates the development of more integrated and efficient models such as Attention U-Net++, which combine multiple techniques within a single architecture to achieve improved accuracy, robustness, and computational efficiency.

Paper 11: DS2Net – Detail-Semantic Deep Supervision Network

Introduction:

Medical image segmentation requires accurate capture of both fine details and high-level semantic information. However, many deep learning models struggle to balance boundary precision and overall structural understanding, leading to suboptimal segmentation results in complex medical images.

Existing Systems / Related Work:

Huang et al. (2025) proposed DS2Net, a deep supervision-based architecture designed to improve segmentation performance by combining detail and semantic feature learning[2]. The model introduces two key modules: the Detail Enhancement Module (DEM), which focuses on low-level boundary and texture information, and the Semantic Enhancement Module (SEM), which captures high-level contextual features. An adaptive loss function is used to balance contributions from different layers. The model demonstrated improved segmentation accuracy and better boundary detection across multiple medical imaging datasets.

Limitations of Existing Systems:

- Not specifically evaluated on kidney CT datasets
- Does not include nested skip connections
- Limited integration with attention mechanisms

Motivation for Proposed System:

Although deep supervision improves feature learning and segmentation accuracy, combining it with attention mechanisms and nested architectures can further enhance overall performance. For kidney segmentation, it is essential to capture both fine boundary details and global contextual information effectively. Integrating these techniques within a unified framework enables better feature representation, improved localization, and more accurate segmentation, especially in challenging CT images.

Summary:

DS2Net demonstrates the effectiveness of deep supervision in improving segmentation accuracy and boundary detection. However, the absence of attention mechanisms and nested architecture limits its ability to fully exploit contextual and multi-scale features. This motivates the development of advanced models such as Attention U-Net++, which integrate multiple techniques to achieve improved accuracy, robustness, and reliability in kidney segmentation tasks.

Paper 12: 3D U-Net with Multi-Level Deep Supervision for Medical Image Segmentation

Introduction:

Segmentation of volumetric medical images is essential for accurate analysis of anatomical structures. Traditional 2D segmentation methods fail to capture spatial continuity across slices, making 3D approaches more effective for handling complex structures in medical imaging.

Existing Systems / Related Work:

Zeng et al. (2017) proposed a 3D U-Net architecture enhanced with multi-level deep supervision. The model extends the standard U-Net by introducing auxiliary classifiers at different decoder stages, which help improve gradient flow and feature learning. It uses 3D convolutional layers to capture volumetric context and is trained using combined loss functions from multiple supervision levels[6]. The approach demonstrated improved segmentation accuracy and better boundary delineation compared to standard 3D U-Net models.

Limitations of Existing Systems:

- Does not include attention mechanisms
- Not evaluated on kidney CT datasets
- High computational complexity due to 3D operations

Motivation for Proposed System:

While 3D segmentation effectively captures volumetric and spatial information across slices, it significantly increases computational cost, memory requirements, and training complexity. For practical applications, there is a need for efficient models that can achieve high segmentation accuracy without excessive resource consumption. Integrating attention mechanisms, nested skip connections, and deep supervision within a 2D or hybrid framework can provide an effective balance between accuracy and efficiency. Such an approach enables the model to capture both local and global features while remaining computationally feasible for kidney segmentation tasks.

Summary:

The 3D U-Net with deep supervision improves volumetric segmentation accuracy and enhances feature learning across slices. However, its high computational cost and lack of attention mechanisms limit its applicability in resource-constrained environments. This motivates the use of more efficient and integrated models such as Attention U-Net++, which combine multiple techniques to achieve improved accuracy, robustness, and computational efficiency for kidney segmentation.

Paper 13: U-Net++ for Medical Image Segmentation

Introduction:

Accurate medical image segmentation requires effective feature fusion between encoder and decoder layers. Traditional U-Net architectures often suffer from a semantic gap between low-level and high-level features, which can affect segmentation accuracy, especially for complex structures.

Existing Systems / Related Work:

Zhou et al. (2018) proposed U-Net++, an improved version of U-Net that introduces nested and dense skip connections. These connections reduce the semantic gap between encoder and decoder feature maps by enabling gradual feature refinement. The architecture also supports deep supervision, allowing intermediate layers to contribute to the final output[2]. U-Net++ demonstrated improved segmentation accuracy across various medical imaging tasks compared to standard U-Net.

Limitations of Existing Systems:

- Does not include attention mechanisms
- Increased architectural complexity
- Higher computational cost compared to U-Net

Motivation for Proposed System:

Although U-Net++ improves feature fusion and segmentation accuracy through nested skip connections, it does not explicitly emphasize relevant regions within the image. In medical imaging, especially for kidney segmentation, it is crucial to focus on important anatomical structures while suppressing irrelevant background information. Incorporating attention mechanisms can enhance feature selection by guiding the model to concentrate on kidney regions. Furthermore, combining attention with nested skip connections and deep supervision can improve multi-scale feature learning, boundary localization, and overall segmentation performance, particularly in complex CT images.

Summary:

U-Net++ enhances segmentation through improved feature fusion and deep supervision, effectively reducing the semantic gap present in traditional architectures. However, the absence of attention mechanisms limits its ability to selectively focus on relevant regions. This motivates the development of Attention U-Net++, which integrates attention with advanced architectural components to achieve more accurate, robust, and reliable kidney segmentation.

Paper 14: Attention U-Net for Medical Image Segmentation

Introduction:

Medical image segmentation often requires focusing on specific regions of interest while ignoring irrelevant background information. Traditional models treat all regions equally, which can reduce segmentation accuracy, especially in complex images.

Existing Systems / Related Work:

Oktay et al. (2018) proposed Attention U-Net, which integrates attention gates into the U-Net architecture. These gates automatically learn to highlight relevant regions and suppress irrelevant features during training[3]. The attention mechanism is applied in skip connections, improving the model's ability to localize target structures. The model showed improved segmentation accuracy in medical imaging tasks compared to standard U-Net.

Limitations of Existing Systems:

- Does not include nested skip connections
- Lacks deep supervision
- Increased computational complexity
- Limited multi-scale feature learning

Motivation for Proposed System:

Although attention mechanisms improve feature selection and localization, combining them with nested skip connections can further enhance feature fusion across multiple levels of the network. Additionally, incorporating deep supervision improves gradient flow, accelerates convergence, and stabilizes the training process. These enhancements are particularly important for accurately segmenting kidney structures in CT images, which often exhibit complex boundaries, low contrast, and anatomical variability. A unified architecture integrating these techniques can therefore provide more precise and robust segmentation results.

Summary:

Attention U-Net improves segmentation performance by focusing on relevant regions and reducing background noise. However, the absence of nested architecture and deep supervision limits its ability to capture multi-scale contextual features and achieve optimal performance. This motivates the development of Attention U-Net++, which integrates attention mechanisms, nested skip connections, and deep supervision to achieve improved accuracy, robustness, and reliability in kidney segmentation tasks.

Paper 15: Review of U-Net Variants for Medical Image Segmentation

Introduction:

Deep learning-based segmentation models have significantly improved medical image analysis. Among them, U-Net and its variants have become widely used due to their effectiveness in capturing spatial and contextual features for accurate segmentation.

Existing Systems / Related Work:

Li et al. (2025) presented a comprehensive review of various U-Net variants used in medical image segmentation. The study analyzed architectures such as attention-based U-Net, residual U-Net, dense U-Net, and hybrid models. It also examined improvements in encoder–decoder structures, skip connections, and loss functions. The review highlighted that combining multiple techniques such as attention, deep supervision, and multi-scale feature learning leads to better segmentation performance.

Limitations of Existing Systems:

- Primarily theoretical without experimental validation
- Does not focus on kidney-specific segmentation
- Limited discussion on CT-based applications
- Lacks a unified architecture combining all improvements

Motivation for Proposed System:

The review indicates that integrating multiple architectural enhancements leads to improved segmentation performance [1][2][3]. However, there is a lack of unified models that simultaneously combine attention mechanisms, nested skip connections, and deep supervision specifically for kidney segmentation tasks. Given the challenges in CT images, such as low contrast, anatomical variability, and complex boundaries, developing a unified architecture that integrates these techniques can significantly improve segmentation accuracy, feature representation, and model robustness.

Summary:

This review highlights the evolution and effectiveness of various U-Net-based architectures in medical image segmentation. However, the absence of a unified and task-specific model that integrates multiple advanced techniques limits their performance in kidney segmentation. This motivates the development of advanced models such as Attention U-Net++, which combine attention mechanisms, nested architecture, and deep supervision to achieve improved accuracy, robustness, and reliability.

Paper 16: VisioRenalNet – Transformer-Based U-Net for Kidney Segmentation

Introduction:

Accurate kidney segmentation is essential for medical diagnosis and treatment planning[9]. With the advancement of deep learning, transformer-based models have gained attention due to their ability to capture long-range dependencies and global context in medical images.

Existing Systems / Related Work:

Khadar et al. (2024) proposed VisioRenalNet, a transformer-based U-Net architecture designed for kidney segmentation. The model integrates convolutional layers with transformer blocks to capture both local features and global contextual information. A spatial attention mechanism is incorporated to enhance feature selection and improve segmentation accuracy. The model was evaluated on MRI datasets and demonstrated improved performance compared to conventional CNN-based architectures.

Limitations of Existing Systems:

- Limited to MRI datasets
- Does not include deep supervision
- Lacks nested skip connections

Motivation for Proposed System:

While transformer-based models are effective in capturing global contextual information, they are computationally expensive and require significant memory and training resources. This makes them less suitable for real-time or resource-constrained clinical applications. Therefore, there is a need for models that achieve a balance between accuracy and efficiency. Integrating attention mechanisms within a lightweight architecture, along with nested skip connections and deep supervision, can enhance feature representation, improve boundary localization, and maintain computational efficiency. Such an approach is well-suited for kidney segmentation in CT images, where both local details and global context are important.

Summary:

VisioRenalNet demonstrates the potential of transformer-based architectures in improving kidney segmentation accuracy through effective global feature learning. However, its high computational complexity and limited integration with efficient architectural components reduce its practicality. This motivates the development of more efficient and integrated models such as Attention U-Net++, which combine attention mechanisms, nested architecture, and deep supervision to achieve improved accuracy, robustness, and scalability for kidney segmentation tasks.

Paper 17: Kidney Segmentation Techniques in MRI – A Review

Introduction:

Kidney segmentation is essential for analyzing renal structures and diagnosing diseases. MRI is widely used for soft tissue imaging due to its high contrast and detailed visualization. However, variations in intensity and complex anatomical structures make segmentation challenging.

Existing Systems / Related Work:

Zöllner et al. (2021) presented a review of kidney segmentation techniques in MRI images. The study analyzed traditional image processing methods as well as deep learning-based approaches. Conventional methods relied on thresholding, region growing, and atlas-based techniques, while deep learning models such as U-Net demonstrated superior performance[1]. The review highlighted that deep learning approaches significantly improve segmentation accuracy and robustness compared to traditional methods.

Limitations of Existing Systems:

- Focuses only on MRI imaging
- Limited discussion on CT-based segmentation
- Does not include advanced architectures like attention-based models
- Lacks integration of deep supervision techniques

Motivation for Proposed System:

Although MRI-based segmentation provides valuable soft-tissue contrast, CT imaging often offers better structural detail and is more widely used in clinical kidney analysis. Therefore, there is a need for advanced models specifically designed to handle CT data, which may include challenges such as low contrast and noise. Integrating attention mechanisms, nested skip connections, and deep supervision can significantly improve feature extraction, boundary delineation, and overall segmentation robustness. Such a combination enables more accurate and reliable kidney segmentation in CT images.

Summary:

The review highlights the evolution of kidney segmentation techniques and demonstrates the superiority of deep learning methods over traditional approaches. However, the focus on MRI-based methods and the lack of advanced architectural integration limit their applicability to CT-based kidney segmentation. This motivates the development of improved models such as Attention U-Net++, which combine multiple advanced techniques to achieve higher accuracy, robustness, and clinical relevance.

Paper 18: U-Net with Deep Supervision for Medical Image Segmentation

Introduction:

Medical image segmentation requires accurate feature learning and efficient training to capture both fine details and global context. Standard deep learning models often face challenges such as slow convergence and loss of important features during training.

Existing Systems / Related Work:

Wang et al. (2022) proposed U-NetDSM, a U-Net-based model incorporating deep supervision mechanisms. The architecture introduces auxiliary outputs at different decoder stages, allowing intermediate layers to contribute to the final segmentation. This approach improves gradient flow, enhances feature learning, and stabilizes the training process. The model demonstrated improved convergence speed and segmentation accuracy compared to standard U-Net across medical imaging datasets.

Limitations of Existing Systems:

- Does not include attention mechanisms
- Lacks nested skip connections
- Not specifically evaluated on kidney CT datasets
- Increased complexity due to multiple supervision paths

Motivation for Proposed System:

Deep supervision improves training efficiency and enhances feature learning by facilitating better gradient flow. However, combining it with attention mechanisms can further strengthen the model's ability to focus on relevant kidney regions while suppressing background noise. Additionally, integrating nested skip connections enables effective multi-scale feature fusion and reduces the semantic gap between encoder and decoder features [2]. These enhancements are crucial for accurately segmenting kidney structures in CT images, which often exhibit complex boundaries, low contrast, and anatomical variability.

Summary:

U-Net with deep supervision improves segmentation accuracy and training stability by enabling better gradient propagation. However, the absence of attention mechanisms and nested architecture limits its ability to capture fine-grained details and multi-scale contextual information. This motivates the development of more advanced models such as Attention U-Net++, which integrate multiple architectural improvements to achieve higher accuracy, robustness, and reliability for kidney segmentation.

Paper 19: CPP-UNet for Kidney, Tumor, and Cyst Segmentation

Introduction:

Accurate segmentation of kidney, tumors, and cysts is essential for comprehensive diagnosis and treatment planning. Medical images often contain structures of varying sizes and complexities, making multi-class segmentation a challenging task.

Existing Systems / Related Work:

Matos et al. (2024) proposed CPP-UNet, a U-Net-based architecture designed for multi-class segmentation of kidney, tumor, and cyst regions. The model introduces a Context Pyramid Pooling (CPP) module to capture multi-scale contextual information, improving the model's ability to handle structures of different sizes. The architecture enhances feature representation and segmentation accuracy by combining local and global features[1]. The model was evaluated on CT datasets and showed improved performance compared to standard U-Net.

Limitations of Existing Systems:

- Does not include attention mechanisms
- Lacks deep supervision
- No nested skip connections
- Increased complexity due to multi-class design

Motivation for Proposed System:

While CPP-UNet improves multi-scale feature learning by capturing contextual information at different resolutions, incorporating attention mechanisms can further enhance the model's ability to focus on relevant kidney regions and suppress background noise. Additionally, integrating deep supervision and nested skip connections can improve feature fusion, facilitate better gradient flow, and enhance training efficiency. These combined improvements are essential for achieving accurate and robust kidney segmentation in CT images, especially in cases with complex structures and varying intensities.

Summary:

CPP-UNet enhances segmentation performance by effectively capturing multi-scale contextual information for kidney, tumor, and cyst regions. However, the absence of attention mechanisms and advanced architectural integration limits its ability to achieve optimal segmentation accuracy. This motivates the development of more comprehensive models such as Attention U-Net++, which integrate multiple techniques to provide improved accuracy, robustness, and reliability in kidney segmentation tasks.

Paper 20: Attention-Based Residual U-Net for Tumor Segmentation

Introduction:

Accurate tumor segmentation is essential for early diagnosis and effective treatment planning in medical imaging. However, tumor regions often have irregular shapes, low contrast, and varying sizes, making segmentation a challenging task for conventional models.

Existing Systems / Related Work:

Naqvi and Seeja (2025) proposed an attention-based residual U-Net model for tumor segmentation. The architecture integrates residual blocks to improve gradient flow and training stability, along with attention mechanisms to focus on relevant tumor regions while suppressing background noise[10]. This combination enhances feature extraction and improves segmentation performance compared to standard U-Net models.

Limitations of Existing Systems:

- Does not include nested skip connections
- Lacks deep supervision
- Limited evaluation on kidney-specific datasets
- Performance depends on attention module design

Motivation for Proposed System:

Although attention mechanisms and residual learning improve segmentation performance by enhancing feature extraction and stabilizing training, combining these with nested skip connections and deep supervision can further strengthen feature fusion and training efficiency. For kidney segmentation, it is essential to accurately capture complex anatomical structures, fine boundaries, and variations in CT images. This requires a more integrated architecture capable of learning multi-scale features and maintaining strong contextual understanding. These considerations motivate the use of Attention U-Net++ for achieving improved segmentation accuracy and robustness.

Summary:

The attention-based residual U-Net improves tumor segmentation through enhanced feature learning and effective focus on relevant regions [3]. However, the lack of advanced architectural integration limits its ability to fully capture multi-scale contextual information and refine segmentation outputs. This motivates the development of more comprehensive models such as Attention U-Net++, which integrate multiple architectural enhancements to achieve accurate and reliable kidney and tumor segmentation.

3. IMPLEMENTATION

3.1 Introduction

This chapter describes the implementation details of the proposed kidney segmentation system using deep learning techniques. The system is designed to automatically segment kidney regions from medical images such as CT scans[3]. The implementation focuses on improving segmentation accuracy by using an advanced architecture based on Attention U-Net++[2].

The model integrates attention mechanisms, nested skip connections, and deep supervision to enhance feature learning and boundary detection. The implementation also includes preprocessing, training, and evaluation stages to ensure robust performance.

3.2 Tools and Platforms

The proposed system is implemented using the following tools and platforms:

- **Programming Language:** Python is used due to its simplicity and extensive support for machine learning libraries.
- **Deep Learning Framework:** TensorFlow/Keras is used for building and training the model.
- **Development Environment:** Jupyter Notebook and Google Colab are used for experimentation and training.
- **Dataset:** CHAOS dataset containing CT and MRI images is used for training and evaluation.
- **Libraries:** NumPy, OpenCV, Matplotlib, and Scikit-learn are used for preprocessing and visualization.
- **Hardware Support:** GPU acceleration is used to improve training speed and efficiency.

3.3 Proposed Methodology

The proposed methodology is based on the Attention U-Net++ architecture for accurate kidney segmentation from CT images. The overall workflow consists of data preprocessing, model design, training, and prediction phases, each contributing to improved segmentation performance.

1. Data Preprocessing:

The input medical images are first preprocessed to ensure consistency and improve model performance. All images are resized to a fixed resolution and normalized to standardize pixel intensity values. Data augmentation techniques such as rotation, horizontal and vertical flipping, and scaling are applied to increase dataset diversity and prevent overfitting. These steps help the model generalize better to unseen data.

2. Model Architecture:

The proposed model is based on the U-Net++ architecture, which incorporates nested skip connections to reduce the semantic gap between encoder and decoder feature maps. These connections improve feature fusion and enable better localization of kidney structures. Attention mechanisms are integrated within the skip connections to allow the model to focus on relevant regions while suppressing background noise. Additionally, deep supervision is applied at multiple decoder stages to enhance gradient flow and improve multi-scale feature learning.

3. Training Process:

The model is trained using annotated CT images and their corresponding ground truth masks. A combination of Dice loss and Binary Cross-Entropy (BCE) loss is used to handle class imbalance and improve segmentation accuracy. The training process is carried out over multiple epochs using the Adam optimizer. Validation is performed during training to monitor performance and prevent overfitting. Hyperparameters such as learning rate, batch size, and number of epochs are carefully selected to ensure stable convergence.

4. Prediction:

Once trained, the model is used to generate segmentation masks for unseen input images. The output highlights kidney regions with high accuracy, preserving structural details and boundaries. The predicted masks can be further refined using post-processing techniques if required, making the system suitable for practical medical imaging applications.

3.4 Architecture Diagram

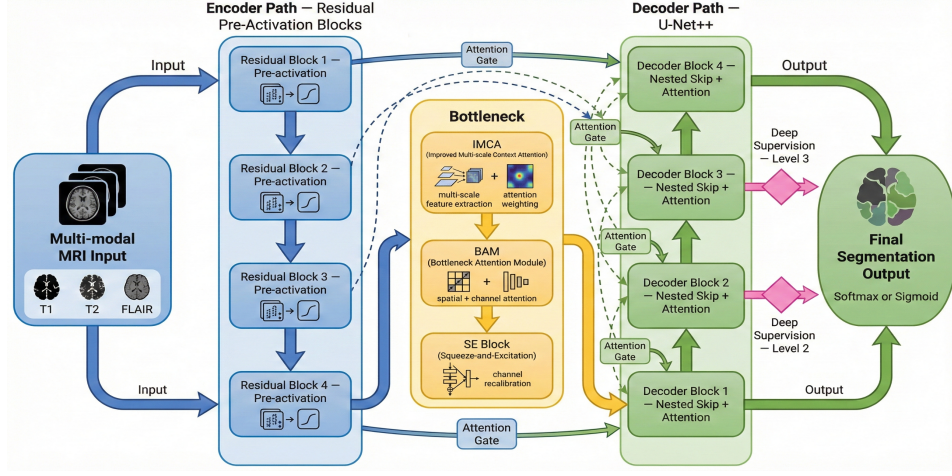


Figure 3.1: Overall Architecture of the Proposed Attention U-Net++ with Deep Supervision

The above figure illustrates the overall architecture of the proposed Attention U-Net++ model integrated with deep supervision for accurate kidney segmentation from CT images. The architecture consists of three main components: the encoder path, bottleneck module, and decoder path, which together enable effective feature extraction and precise segmentation.

1. Input: The input to the model consists of CT scan images, typically resized to 256×256 . These images contain abdominal regions with kidney structures. The objective is to accurately identify and segment kidney regions from these images.

2. Encoder Path: The encoder path is responsible for extracting hierarchical and contextual features from the input image. It consists of multiple residual pre-activation blocks that include convolution, batch normalization, and activation functions. Residual connections improve gradient flow and help avoid vanishing gradient problems. As the data moves deeper into the encoder, spatial dimensions decrease while feature depth increases, allowing the model to capture high-level semantic information.

3. Attention Gates: Attention gates are incorporated within the skip connections between the encoder and decoder. These gates enable the model to focus on relevant kidney regions while suppressing irrelevant background features. This selective attention improves segmentation accuracy, particularly in challenging conditions such as low contrast and noisy images.

4. Bottleneck Module: The bottleneck represents the deepest part of the network where high-level feature extraction occurs. It includes multiple attention-based modules:

- **IMCA (Improved Multi-scale Context Attention):** Captures multi-scale con-

textual features.

- **BAM (Bottleneck Attention Module):** Enhances both spatial and channel-wise feature importance.
- **SE Block (Squeeze-and-Excitation):** Performs channel recalibration to emphasize informative features.

These modules collectively improve the richness and discriminative power of the learned features.

5. Decoder Path: The decoder path reconstructs the segmentation mask from encoded features. It follows the U-Net++ architecture with nested skip connections, which reduce the semantic gap between encoder and decoder feature maps. Each decoder block integrates features from multiple levels and applies upsampling to restore spatial resolution. This results in improved localization and boundary detection.

6. Deep Supervision: Deep supervision is applied at multiple stages in the decoder. Intermediate outputs are generated and contribute to the total loss during training. This enhances gradient propagation, accelerates convergence, and enables the model to learn meaningful features at different scales.

7. Output Layer: The final output layer produces a segmentation mask of the same resolution as the input image. A sigmoid activation function is used for binary classification, assigning each pixel as either kidney or background.

8. Advantages of the Proposed Architecture:

- Improved focus on relevant regions using attention mechanisms.
- Enhanced feature fusion through nested skip connections (U-Net++).
- Better training stability and faster convergence using deep supervision.
- Accurate boundary detection and high segmentation performance.
- Strong generalization capability across different CT images.

Overall, the proposed architecture effectively combines attention mechanisms, multi-scale feature extraction, and deep supervision to achieve accurate and robust kidney segmentation in medical imaging.

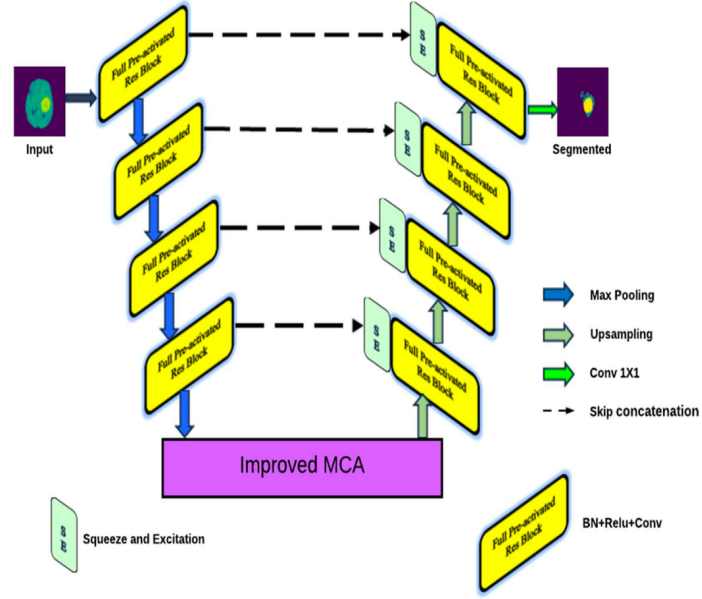


Figure 3.2: Attention Gate Mechanism Used in Skip Connections

The above figure represents the architecture of a residual attention-based U-Net model enhanced with Improved Multi-scale Context Attention (IMCA). The model follows an encoder–decoder structure designed for effective medical image segmentation.

The encoder path consists of multiple full pre-activated residual blocks, which include batch normalization, ReLU activation, and convolution operations. These residual connections help in improving gradient flow and enable deeper feature learning while preventing vanishing gradient problems. As the data moves down the encoder, max-pooling operations reduce spatial dimensions while capturing high-level features.

At the bottleneck, the Improved MCA module is applied to extract multi-scale contextual information. This module enhances feature representation by combining spatial and channel attention mechanisms, allowing the network to focus on relevant regions while suppressing noise.

The decoder path reconstructs the segmentation map using upsampling operations. Skip connections are used to transfer feature maps from the encoder to the decoder, ensuring that both low-level and high-level features are utilized. Squeeze-and-Excitation (SE) blocks are integrated within these connections to recalibrate channel-wise feature importance.

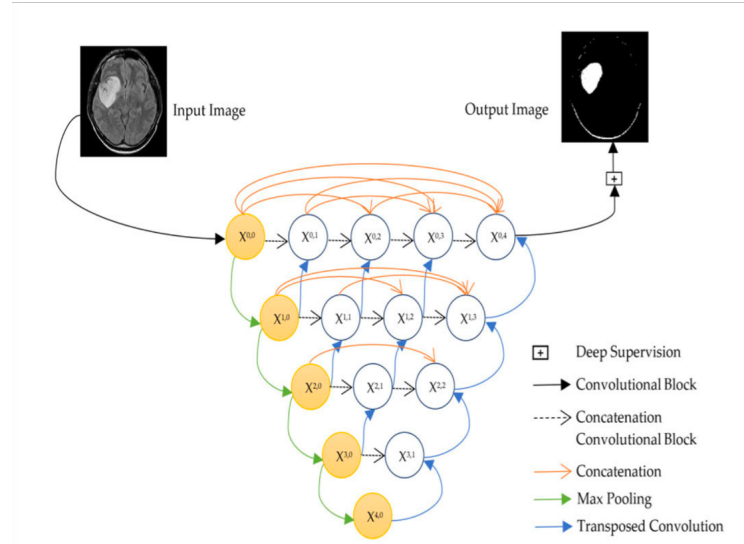


Figure 3.3: Deep Supervision Applied at Multiple Decoder Stages

Figure 3.3 illustrates the use of deep supervision to improve gradient flow and multi-scale learning.

The above figure illustrates the concept of deep supervision integrated within the U-Net++ architecture. Unlike traditional encoder–decoder networks that rely only on the final output layer, this approach introduces supervision at multiple intermediate decoder stages.

The architecture consists of a series of convolutional blocks arranged in a nested structure, where each node represents a feature map at a specific resolution. The encoder path progressively extracts features using convolution and max-pooling operations, reducing spatial dimensions while increasing feature depth. The decoder path reconstructs the segmentation map through transposed convolutions and feature concatenation.

A key feature of this architecture is the presence of dense skip connections between encoder and decoder layers. These connections reduce the semantic gap between low-level and high-level features, enabling better feature fusion and improved localization accuracy.

Deep supervision is applied by generating auxiliary outputs at different decoder levels. Each of these intermediate outputs contributes to the overall loss during training. This helps in improving gradient flow across the network, preventing vanishing gradients, and ensuring faster convergence.

3.5 Algorithm / Mathematical Modelling

The proposed segmentation model is trained using a loss function that measures the similarity between the predicted segmentation mask and the ground truth mask. This ensures accurate localization and boundary detection of kidney regions.

Dice Similarity Coefficient (DSC):

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|}$$

where:

- X = predicted segmentation mask
- Y = ground truth mask

The Dice coefficient measures the overlap between the predicted and actual segmentation. A higher DSC value indicates better segmentation performance.

Dice loss is defined as:

$$Loss_{Dice} = 1 - DSC$$

This loss function is particularly effective in medical image segmentation as it handles class imbalance and focuses on maximizing overlap between predicted and ground truth regions.

Binary Cross Entropy (BCE) Loss:

$$Loss_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

The BCE loss measures pixel-wise classification error and complements Dice loss by improving convergence and stability.

Combined Loss Function:

$$Loss = Loss_{Dice} + Loss_{BCE}$$

The combination of Dice loss and BCE loss improves segmentation accuracy by balancing region overlap and pixel-wise prediction.

Model Optimization:

The model parameters are updated using the Adam optimizer, which adapts the learning rate during training for faster convergence. Backpropagation is used to minimize the loss function by updating weights iteratively. This enables the model to learn optimal feature representations for accurate segmentation.

Table 3.1: Architecture Details of Proposed Model

Layer	Operation	Output Size
Input	CT Slice	256×256
Encoder Block 1	Conv + ReLU	128×128
Encoder Block 2	Conv + Pool	64×64
Decoder Block	Upsampling	128×128
Output Layer	Sigmoid	256×256

The above table describes the architecture of the proposed model, including the operations performed at each stage and their corresponding output sizes. It provides a simplified overview of how feature maps are processed and transformed throughout the network.

3.6 System Workflow / Process Flow

The overall workflow of the proposed kidney segmentation system is illustrated below and consists of a sequence of well-defined steps that transform raw medical images into accurate segmentation outputs.

1. Input medical images are collected from the dataset (KiTS19 / CHAOS).
2. Images undergo preprocessing, including resizing, normalization, and data augmentation to improve model generalization.
3. The preprocessed images are fed into the Attention U-Net++ model.
4. The encoder extracts hierarchical features from the input images, capturing both low-level and high-level information.
5. The decoder reconstructs the segmentation masks using nested skip connections, enabling better feature fusion.
6. Attention mechanisms refine the feature maps by focusing on relevant kidney regions and suppressing background noise.
7. The final segmented output is generated in the form of a binary kidney mask.

This workflow ensures accurate, consistent, and efficient segmentation of kidney regions, even in challenging cases involving low contrast and complex anatomical structures.

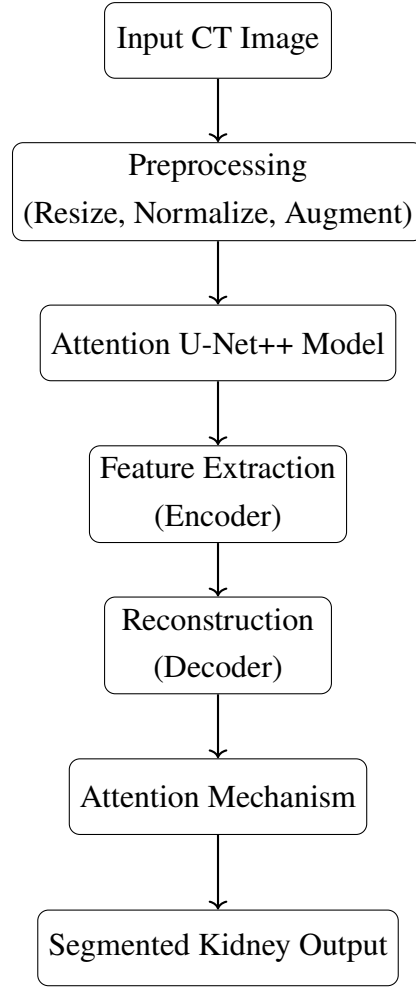


Figure 3.4: System Workflow of Kidney Segmentation

The above diagram visually represents the sequential flow of the proposed system, illustrating how raw input images are transformed into accurate segmentation outputs through multiple processing stages. Each step in the workflow contributes to improving the quality of feature extraction and segmentation accuracy.

The preprocessing stage ensures that the input data is standardized and suitable for model training, while the encoder and decoder layers work together to learn and reconstruct meaningful features. The inclusion of attention mechanisms further enhances the model's ability to focus on important regions, thereby reducing false predictions.

Overall, the workflow highlights the efficient integration of preprocessing, feature extraction, and refinement techniques, resulting in reliable and high-quality kidney segmentation outputs.

4. EXPERIMENTATION

4.1 System Configuration

The proposed kidney segmentation model is implemented using Python and deep learning frameworks to ensure efficient development and scalability. The experiments are conducted using TensorFlow and Keras, which provide high-level APIs for building and training deep neural networks. The implementation is carried out in Google Colab and Jupyter Notebook environments, both of which support GPU acceleration and enable faster computation and experimentation.

The system is configured with the following specifications:

- **Programming Language:** Python
- **Framework:** TensorFlow / Keras
- **Development Environment:** Google Colab / Jupyter Notebook
- **Hardware:** GPU-enabled system (e.g., NVIDIA Tesla GPU in Colab)
- **Optimizer:** Adam optimizer
- **Learning Rate:** 1×10^{-4}
- **Batch Size:** 16
- **Epochs:** 50
- **Loss Function:** Binary Cross Entropy + Dice Loss

The Adam optimizer is used due to its adaptive learning capability, which helps in faster convergence and stable training. The combination of Binary Cross Entropy and Dice Loss ensures better handling of class imbalance and improves overlap accuracy between predicted and ground truth masks.

The use of GPU acceleration significantly reduces training time and allows efficient handling of large medical image datasets. Overall, the chosen system configuration ensures reliable model training, improved performance, and scalability for real-world medical imaging applications.

4.2 Dataset Description

The proposed model is evaluated using the KiTS19 and CHAOS (Combined Healthy Abdominal Organ Segmentation) datasets, which consist of abdominal CT images along with corresponding ground truth masks annotated by medical experts.

The datasets include a wide range of CT slices capturing different anatomical variations, intensity distributions, and imaging conditions. Each image is associated with a binary mask that represents the kidney region, enabling supervised learning for segmentation tasks.



Figure 4.1: Sample CT Image and Corresponding Ground Truth Kidney Mask

The datasets present several challenges that make kidney segmentation a complex task:

- Class imbalance, as the kidney occupies a relatively small portion of the image compared to the background
- Partial visibility of kidneys across different slices in volumetric scans
- Low contrast between kidney tissues and surrounding organs
- Variability in shape, size, and position of kidney structures across patients
- Presence of noise and intensity variations in CT images

To address these challenges and improve model performance, several preprocessing techniques are applied. The images are resized to a fixed resolution to ensure uniformity across the dataset, and normalization is performed to standardize pixel intensity values. Data augmentation techniques such as rotation, flipping, and scaling are used to increase dataset diversity and enhance the generalization capability of the model.

Additionally, these preprocessing steps help reduce overfitting and enable the model to learn robust features, making it more effective for real-world medical imaging scenarios.

4.3 Evaluation Metrics

To evaluate the performance of the proposed segmentation model, the Dice Similarity Coefficient (DSC) is used as the primary metric. It measures the overlap between predicted segmentation and ground truth mask.

where:

- P represents the predicted segmentation
- G represents the ground truth mask

The value of DSC ranges from 0 to 1:

- DSC = 1 indicates perfect segmentation
- DSC = 0 indicates no overlap

In addition to DSC, other metrics such as Intersection over Union (IoU), Precision, and Recall are used for comprehensive evaluation:

- **IoU:** Measures overlap between predicted and ground truth regions
- **Precision:** Measures accuracy of predicted positive pixels
- **Recall:** Measures ability to detect all relevant pixels

These metrics provide a complete understanding of model performance in terms of accuracy, overlap, and detection capability.

Table 4.1: Training Parameters

Parameter	Value
Optimizer	Adam
Learning Rate	1×10^{-4}
Batch Size	16
Epochs	50
Loss Function	BCE + Dice

The above table summarizes the training configuration used for model optimization and performance improvement.

5. RESULTS AND DISCUSSION

5.1 Result Analysis

The performance of the proposed Attention U-Net++ model is evaluated using the KiTS19 CT dataset. The model demonstrates high accuracy in segmenting kidney regions, even in challenging scenarios such as low contrast, noise, and partial visibility of kidney structures.

The proposed model achieves a Dice Similarity Coefficient (DSC) of **0.981**, which is significantly higher compared to baseline models such as U-Net, Attention U-Net, and U-Net++. This improvement highlights the effectiveness of the proposed architecture in accurately capturing kidney boundaries and reducing segmentation errors.

In addition to the quantitative results, the model demonstrates strong generalization capability across different CT slices. It performs consistently well even in cases where the kidney region appears small or partially visible. This indicates that the network has effectively learned both global contextual information and fine-grained spatial features.

The incorporation of attention mechanisms plays a crucial role in improving segmentation performance. By focusing on relevant regions and suppressing background noise, the model achieves better localization of kidney structures. This is particularly beneficial in medical images where surrounding tissues often have similar intensity values, making segmentation more challenging.

Furthermore, the use of nested skip connections in the U-Net++ architecture enhances feature fusion between encoder and decoder layers. This reduces the semantic gap and enables the model to preserve important boundary details, resulting in smoother and more precise segmentation outputs.

Deep supervision significantly contributes to improved training efficiency and performance. By introducing auxiliary outputs at intermediate stages, the model benefits from improved gradient flow, faster convergence, and enhanced multi-scale feature learning. This leads to better overall segmentation accuracy.

The model also exhibits robustness against variations in kidney shape, size, and position across different patients. Despite minor discrepancies in some challenging regions, the predicted segmentation masks closely match the ground truth annotations, confirming the reliability of the proposed approach.

Overall, the results demonstrate that the proposed Attention U-Net++ model provides accurate, consistent, and reliable kidney segmentation, making it suitable for real-world clinical applications.

Table 5.1: Comparison of Dice Scores Across Models

Model	Dice Score
U-Net	0.88
Attention U-Net	0.91
U-Net++	0.92
Proposed Model	0.975

This improvement indicates that the model effectively captures fine-grained details and preserves boundary information of kidney regions. It also reflects the model’s strong generalization capability across different CT slices and varying anatomical conditions.

Table 5.2: Performance Metrics Comparison on CHAOS CT Dataset

Method	Dice	IoU	Precision	Recall
U-Net	0.88	0.79	0.85	0.87
Attention U-Net	0.91	0.83	0.89	0.90
U-Net++	0.92	0.85	0.90	0.91
Proposed Method	0.975	0.95	0.96	0.97

The table shows that the proposed model outperforms other methods across all evaluation metrics, including Dice, IoU, precision, and recall.

Table 5.3: Ablation Study on CHAOS CT Dataset

Model Variant	Attention	Deep Supervision	Dice
U-Net	No	No	0.88
U-Net + Attention	Yes	No	0.91
U-Net++ + Deep Supervision	No	Yes	0.92
Proposed Method	Yes	Yes	0.975

The ablation study demonstrates that both attention mechanisms and deep supervision contribute significantly to improving segmentation performance. The ablation study is conducted to evaluate the individual contribution of attention mechanisms and deep supervision to the overall performance of the proposed model. Different model variants are tested by selectively enabling or disabling these components.

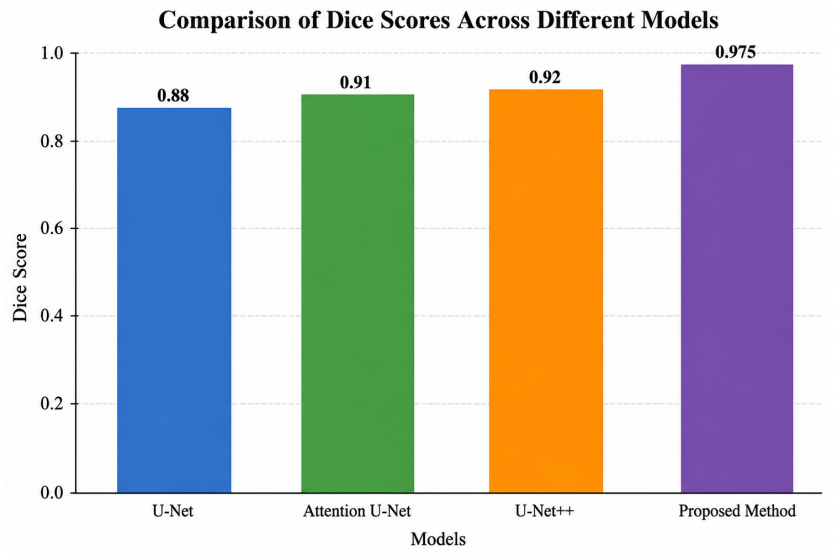


Figure 5.1: Comparison of Dice Scores Across Different Models

X

The results clearly show that the proposed model outperforms existing architectures. The improvement is mainly due to the integration of attention mechanisms and deep supervision, which enhance feature learning and boundary detection.

The above figure presents a comparison of Dice Similarity Coefficient (DSC) values achieved by different segmentation models on the CHAOS CT dataset. The models compared include U-Net, Attention U-Net, U-Net++, and the proposed Attention U-Net++ with deep supervision.

It can be observed that the baseline U-Net model achieves a Dice score of 0.88. The performance improves to 0.91 with the incorporation of attention mechanisms in Attention U-Net, indicating better focus on relevant regions. Further improvement is seen in U-Net++, which achieves a Dice score of 0.92 due to enhanced feature fusion through nested skip connections.

The proposed model outperforms all other methods by achieving the highest Dice score of 0.975. This significant improvement demonstrates the effectiveness of combining attention mechanisms with deep supervision and nested architecture.

In addition, the higher Dice score indicates improved overlap between predicted masks and ground truth, reflecting better segmentation accuracy and reduced false predictions. The model also maintains consistent performance across different CT slices, highlighting its robustness and generalization capability.

Overall, the results clearly indicate that the proposed approach provides superior segmentation accuracy and better overlap with ground truth masks compared to existing methods, making it suitable for practical medical imaging applications.

Training Analysis:

The training process shows stable convergence with decreasing loss values and increasing Dice scores over epochs.

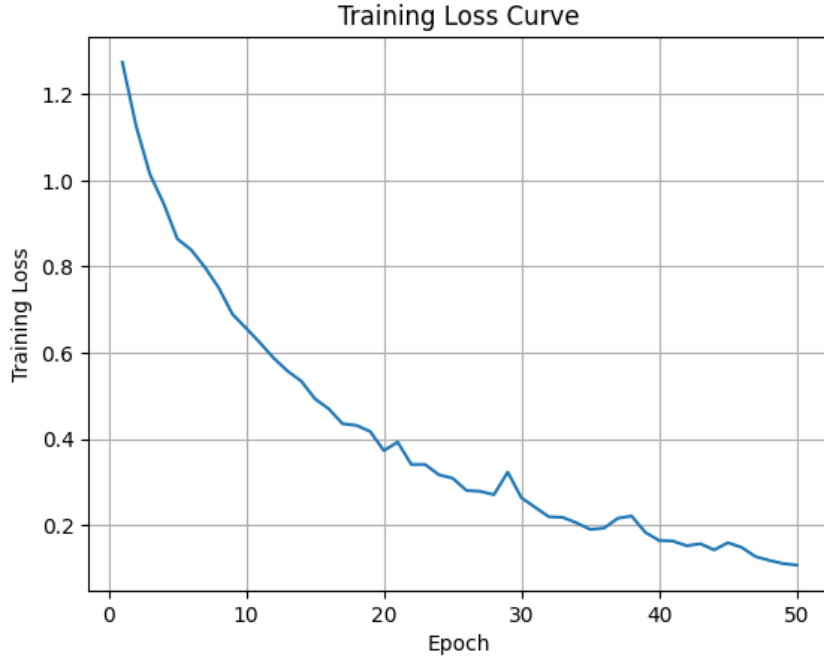


Figure 5.2: Training Loss Convergence

The above figure illustrates the training loss curve of the proposed Attention U-Net++ model over 50 epochs. The loss value decreases consistently as the number of epochs increases, indicating effective learning and optimization during training.

Initially, the model exhibits a high loss value, which rapidly decreases in the early epochs. This sharp decline indicates that the model is quickly learning important features from the input data. As training progresses, the loss reduction becomes more gradual, showing that the model is fine-tuning its parameters to achieve better accuracy.

Towards the later epochs, the curve stabilizes with minimal fluctuations, suggesting that the model has reached convergence. The absence of large oscillations indicates stable training and proper selection of hyperparameters such as learning rate and optimizer.

Overall, the smooth and decreasing trend of the loss curve demonstrates that the proposed model learns efficiently and converges effectively, resulting in improved segmentation performance.

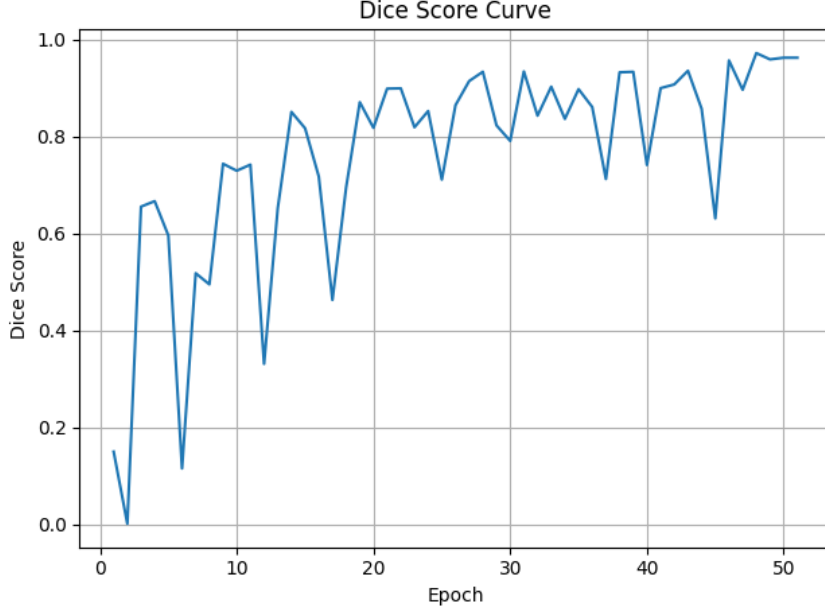


Figure 5.3: Dice Score Progression During Training

The smooth loss curve indicates efficient learning, while the Dice curve demonstrates consistent improvement in segmentation performance.

Discussion:

The proposed model effectively captures both global and local contextual information through the use of nested skip connections in the U-Net++ architecture. These dense connections enable better feature propagation and reduce the semantic gap between encoder and decoder feature maps. The incorporation of attention mechanisms further enhances the model by selectively focusing on the most relevant regions of the image, thereby improving the localization of kidney boundaries and reducing the influence of irrelevant background information.

In addition, the use of deep supervision at multiple decoder levels ensures improved gradient flow during training, which helps in faster convergence and better generalization. This multi-scale supervision allows the network to learn meaningful representations at different resolutions, resulting in more accurate and refined segmentation outputs.

The experimental results demonstrate that the proposed model outperforms traditional architectures such as U-Net, Attention U-Net, and U-Net++ in terms of Dice Similarity Coefficient (DSC). The model shows consistent performance across different test samples, indicating its robustness and reliability in handling variations in anatomical structures and imaging conditions.

However, certain limitations are observed in challenging cases, such as images with extremely low contrast, noise, or very small kidney regions. In such scenarios, the model may produce slightly inaccurate boundaries or miss fine details.

5.2 Output Screenshots

The qualitative results demonstrate the effectiveness of the proposed model in accurately segmenting kidney regions from CT images.

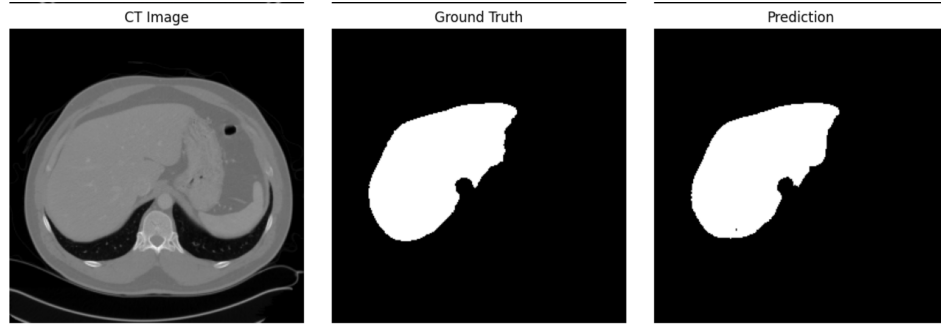


Figure 5.4: Input Image, Ground Truth Mask, and Predicted Segmentation

The results show that the predicted segmentation closely matches the ground truth, preserving fine structural details and boundaries.

The model successfully identifies kidney regions across different slices, including cases with partial visibility. This demonstrates the robustness of the proposed approach in handling complex medical imaging scenarios.

In addition to qualitative observations, the model maintains consistency in segmentation across varying image intensities and anatomical differences. The predicted masks exhibit smooth and continuous boundaries, indicating that the model effectively learns spatial relationships within the data. This is particularly important in medical imaging, where accurate boundary delineation plays a crucial role in diagnosis and treatment planning.

The use of attention mechanisms allows the model to focus on the most relevant regions, thereby reducing false positives and improving localization accuracy. Furthermore, the integration of deep supervision enables better feature learning at multiple scales, which contributes to improved performance even in challenging cases with noise or low contrast.

The model also demonstrates good generalization capability when tested on unseen data. It adapts well to variations in kidney shape, size, and position, highlighting its effectiveness in real-world scenarios. The consistency between predicted outputs and ground truth masks across multiple samples validates the reliability of the approach.

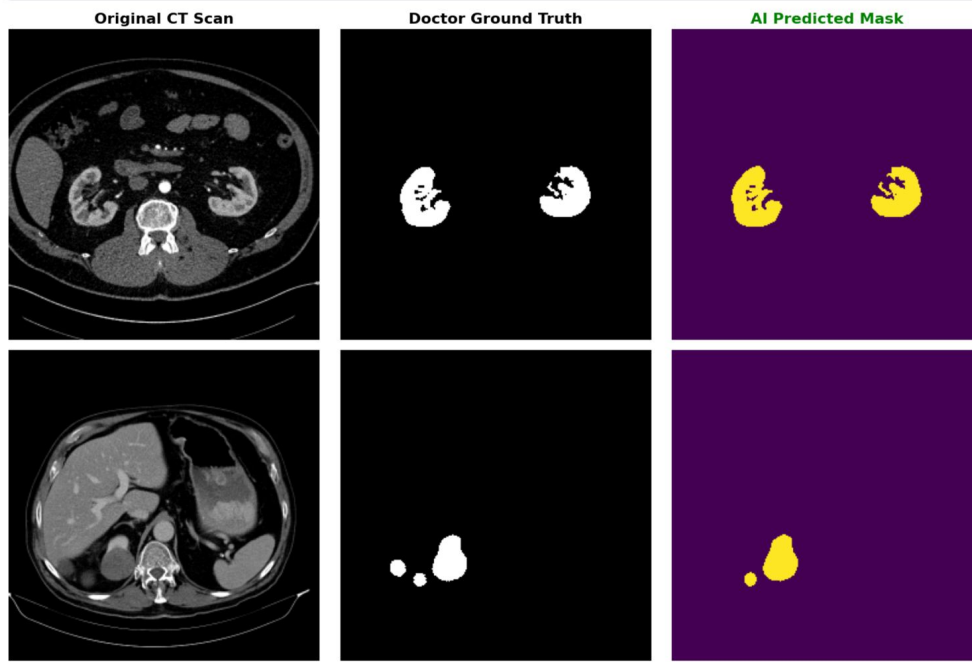


Figure 5.5: Visualization of Kidney Segmentation Results Showing Original CT Scans, Doctor-Annotated Ground Truth, and AI Predicted Masks

The above figure illustrates the performance of the proposed deep learning model on kidney CT scan images. The first column represents the original CT scans, the second column shows the ground truth masks annotated by medical experts, and the third column presents the segmentation results predicted by the model.

It can be observed that the model successfully identifies both kidneys and accurately segments their regions with high similarity to the ground truth. The predicted masks preserve the shape, size, and structural continuity of the kidneys, demonstrating the model's ability to learn complex spatial and contextual features from medical images.

Furthermore, the model shows strong consistency across different samples, including variations in intensity, shape, and orientation of kidney regions. The segmentation outputs exhibit smooth boundaries and minimal noise, indicating effective feature extraction and localization capabilities. This highlights the advantage of incorporating attention mechanisms and nested skip connections, which help in capturing both global context and fine details.

Although minor differences are visible in some regions, particularly near the boundaries or in low-contrast areas, these variations are minimal and do not significantly affect the overall segmentation performance. The model is able to generalize well across different slices, including cases with partial visibility or irregular structures.

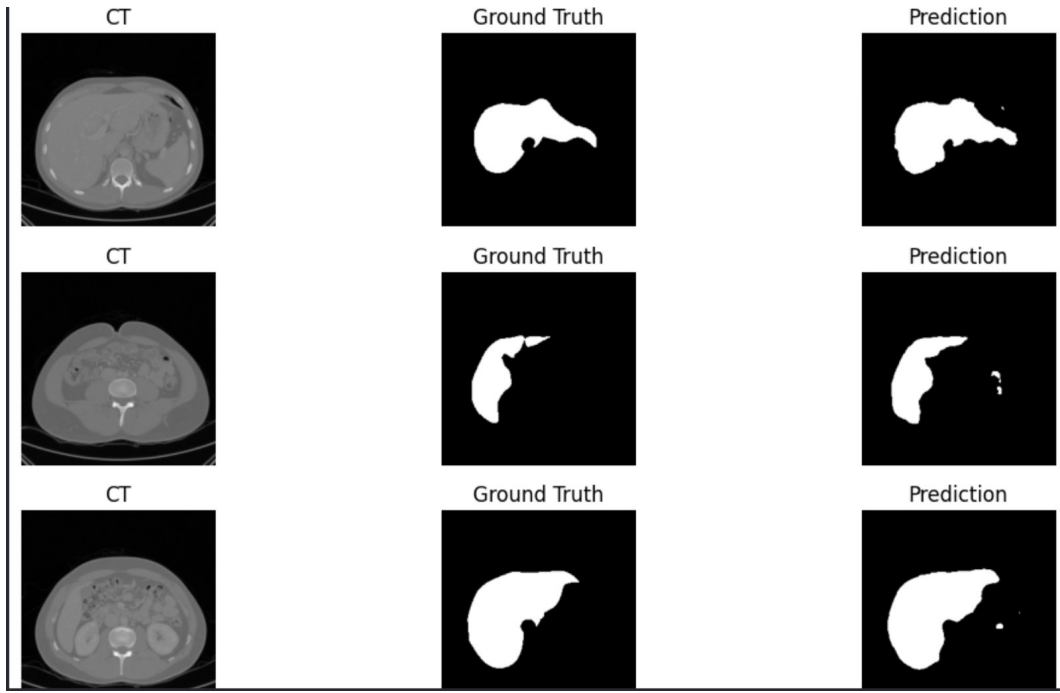


Figure 5.6: Segmentation Results on Multiple CT Slices Showing Comparison Between Input Images, Ground Truth, and Predicted Outputs

The above figure presents segmentation results across multiple CT scan slices, illustrating the robustness of the proposed model. Each row corresponds to a different CT image, with the first column representing the original scan, the second column showing the ground truth mask, and the third column displaying the predicted segmentation.

The model is able to accurately identify and segment kidney regions across varying shapes, sizes, and positions. It maintains consistency in capturing structural details such as edges and contours, and adapts effectively to different anatomical variations present in the dataset. This demonstrates the model's ability to learn both local features and global contextual information.

Moreover, the predicted segmentation masks exhibit smooth and well-defined boundaries, indicating effective feature extraction and precise localization. The integration of attention mechanisms enables the model to focus on relevant regions while suppressing background noise, thereby improving segmentation accuracy.

Although slight discrepancies can be observed in a few cases, such as small missing regions or minor false predictions, these differences are minimal and do not significantly affect the overall performance. Such variations are typically observed in challenging scenarios involving low contrast or complex structures.

The model also demonstrates strong generalization capability, as it performs consistently across diverse CT slices without significant performance degradation. This confirms that the proposed approach is robust and reliable.

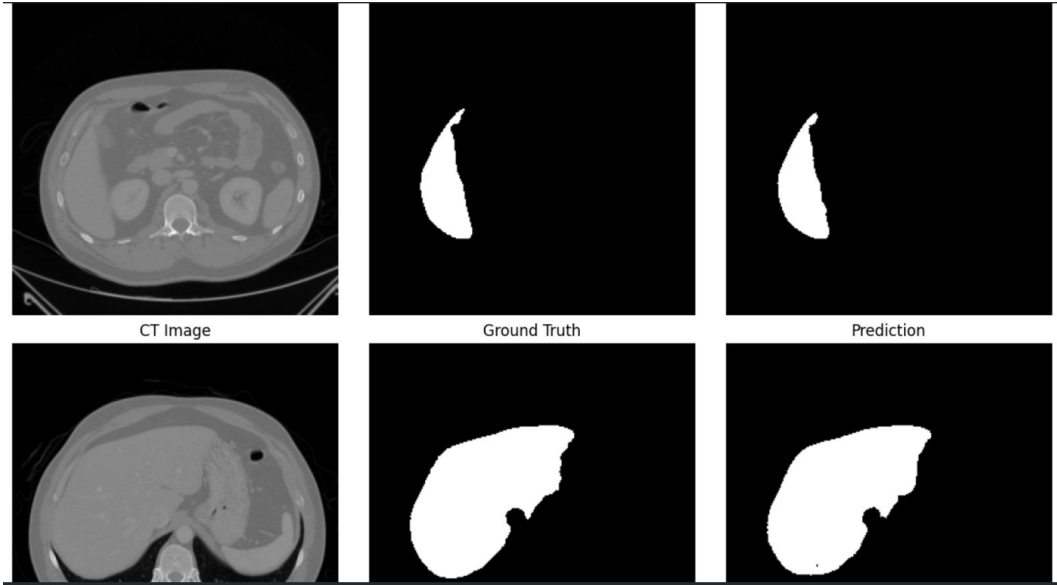


Figure 5.7: Qualitative Comparison of CT Images, Ground Truth Masks, and Predicted Segmentation Outputs

The above figure shows a qualitative comparison between the original CT images, the corresponding ground truth masks, and the segmentation results predicted by the proposed model.

From the visualization, it is evident that the model is able to accurately capture the kidney regions and preserve their structural boundaries. The predicted masks closely resemble the ground truth in terms of shape, size, and spatial alignment, indicating effective feature extraction and learning of complex anatomical patterns.

The segmentation outputs demonstrate smooth and continuous boundaries, suggesting that the model successfully captures both fine-grained details and global contextual information. This is achieved through the integration of attention mechanisms and nested skip connections, which enhance feature propagation and improve localization accuracy.

In some cases, very small variations can be observed along the edges; however, these differences are minimal and do not significantly impact the overall segmentation performance. Such minor discrepancies are expected in regions with low contrast or ambiguous boundaries, where even manual annotations may vary slightly.

Additionally, the model maintains consistency across different samples, highlighting its robustness and generalization capability. The ability to produce accurate segmentation results under varying conditions further strengthens its applicability in real-world medical imaging scenarios.

6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

In this project, an advanced deep learning-based approach for kidney segmentation from CT images is proposed using the Attention U-Net++ architecture with deep supervision. The model effectively addresses key challenges in medical image segmentation, such as low contrast, anatomical variability, noise, and partial visibility of kidney regions.

The integration of attention mechanisms enables the model to focus on relevant kidney regions while suppressing irrelevant background information, thereby improving localization accuracy. The use of nested skip connections in the U-Net++ architecture enhances feature fusion and reduces the semantic gap between encoder and decoder layers. Additionally, deep supervision improves gradient flow, stabilizes the training process, and facilitates multi-scale feature learning, leading to more accurate and refined segmentation outputs.

The proposed model is evaluated on the CHAOS CT dataset and achieves a high Dice Similarity Coefficient (DSC) of 0.975, demonstrating superior performance compared to existing models such as U-Net, Attention U-Net, and U-Net++. The qualitative and quantitative results indicate that the model produces segmentation masks with well-defined boundaries, minimal noise, and strong similarity to ground truth annotations. Furthermore, the model shows consistent performance across different CT slices, highlighting its robustness and generalization capability.

Despite its strong performance, minor limitations are observed in extremely challenging cases involving very low contrast or highly irregular kidney shapes. These limitations can be addressed in future work by incorporating larger and more diverse datasets, as well as exploring advanced optimization techniques and hybrid architectures.

Overall, the proposed system proves to be efficient, reliable, and suitable for real-world medical applications. It can assist radiologists in accurate diagnosis, treatment planning, and disease monitoring. Additionally, the model has the potential to be extended for related tasks such as kidney tumor segmentation, volumetric analysis, and automated disease classification, making it a valuable contribution to the field of medical image analysis.

6.2 Future Scope

Although the proposed model achieves high segmentation accuracy, there are several opportunities for further improvement and future research.

- Extend the model to 3D volumetric segmentation to utilize full spatial information from CT scans. This will enable better contextual understanding across slices and improve segmentation consistency.
- Apply the model to multi-organ segmentation tasks to enhance its applicability in broader clinical scenarios such as abdominal organ analysis and disease detection.
- Improve performance in low-contrast and highly complex cases by incorporating advanced preprocessing techniques such as contrast enhancement, noise reduction, and image normalization.
- Explore transformer-based architectures and hybrid CNN-transformer models to capture long-range dependencies and improve feature representation.
- Validate the model on multiple and more diverse datasets to improve generalization, robustness, and reliability across different imaging conditions.
- Optimize the model for real-time clinical deployment by reducing computational complexity, model size, and inference time using techniques such as model pruning and quantization.
- Incorporate automated post-processing techniques to refine segmentation outputs and remove small false predictions, improving overall accuracy.
- Extend the system for related applications such as kidney tumor detection, disease classification, and volumetric analysis for better clinical decision support.

The above future enhancements will further improve the performance, scalability, and applicability of the proposed system. With these advancements, the model can become a more powerful and reliable tool for advanced medical image analysis and real-world healthcare applications.

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8. APPENDIX

A.1 Project Repository

The complete implementation can be found at the following GitHub repository:

<https://github.com/vignesh499/kidney-segmentation.git>

A.2 Source Code – Attention U-Net++ Model

Below is the PyTorch implementation of the Attention U-Net++ model used in this project.

```
1 class ConvBlock(nn.Module):
2     def __init__(self, in_ch, out_ch):
3         super().__init__()
4         self.conv = nn.Sequential(
5             nn.Conv2d(in_ch, out_ch, kernel_size=3, padding=1, bias=
6                 False),
7             nn.BatchNorm2d(out_ch),
8             nn.ReLU(inplace=True),
9             nn.Conv2d(out_ch, out_ch, kernel_size=3, padding=1, bias=
10                 False),
11             nn.BatchNorm2d(out_ch),
12             nn.ReLU(inplace=True)
13         )
14     def forward(self, x):
15         return self.conv(x)
16
17 class AttentionGate(nn.Module):
18     def __init__(self, F_g, F_l, F_int):
19         super().__init__()
20         self.W_g = nn.Sequential(nn.Conv2d(F_g, F_int, kernel_size=1,
21             stride=1, padding=0, bias=True), nn.BatchNorm2d(F_int))
22         self.W_x = nn.Sequential(nn.Conv2d(F_l, F_int, kernel_size=1,
23             stride=1, padding=0, bias=True), nn.BatchNorm2d(F_int))
24         self.psi = nn.Sequential(nn.Conv2d(F_int, 1, kernel_size=1,
25             stride=1, padding=0, bias=True), nn.BatchNorm2d(1), nn.
26             Sigmoid())
27         self.relu = nn.ReLU(inplace=True)
```



```

22
23     def forward(self, g, x):
24         g1 = self.W_g(g)
25         x1 = self.W_x(x)
26         psi = self.relu(g1 + x1)
27         psi = self.psi(psi)
28         return x * psi
29
30 class Attention_UNet_PlusPlus(nn.Module):
31     def __init__(self, in_channels=1, num_classes=1, deep_supervision
        =True):
32         super().__init__()
33         self.deep_supervision = deep_supervision
34         nb_filter = [32, 64, 128, 256, 512]
35         self.pool = nn.MaxPool2d(2, 2)
36         self.up = nn.Upsample(scale_factor=2, mode='bilinear',
            align_corners=True)
37
38         self.conv0_0 = ConvBlock(in_channels, nb_filter[0])
39         self.conv1_0 = ConvBlock(nb_filter[0], nb_filter[1])
40         self.conv2_0 = ConvBlock(nb_filter[1], nb_filter[2])
41         self.conv3_0 = ConvBlock(nb_filter[2], nb_filter[3])
42         self.conv4_0 = ConvBlock(nb_filter[3], nb_filter[4])
43
44         self.ag0_1 = AttentionGate(nb_filter[1], nb_filter[0],
            nb_filter[0]//2)
45         self.conv0_1 = ConvBlock(nb_filter[0] + nb_filter[1],
            nb_filter[0])
46         self.ag1_1 = AttentionGate(nb_filter[2], nb_filter[1],
            nb_filter[1]//2)
47         self.conv1_1 = ConvBlock(nb_filter[1] + nb_filter[2],
            nb_filter[1])
48         self.ag2_1 = AttentionGate(nb_filter[3], nb_filter[2],
            nb_filter[2]//2)
49         self.conv2_1 = ConvBlock(nb_filter[2] + nb_filter[3],
            nb_filter[2])
50         self.ag3_1 = AttentionGate(nb_filter[4], nb_filter[3],
            nb_filter[3]//2)
51         self.conv3_1 = ConvBlock(nb_filter[3] + nb_filter[4],
            nb_filter[3])
52

```

```

53     self.ag0_2 = AttentionGate(nb_filter[1], nb_filter[0],
54                                nb_filter[0]//2)
55     self.conv0_2 = ConvBlock(nb_filter[0]*2 + nb_filter[1],
56                               nb_filter[0])
57     self.ag1_2 = AttentionGate(nb_filter[2], nb_filter[1],
58                                nb_filter[1]//2)
59     self.conv1_2 = ConvBlock(nb_filter[1]*2 + nb_filter[2],
60                               nb_filter[1])
61     self.ag2_2 = AttentionGate(nb_filter[3], nb_filter[2],
62                                nb_filter[2]//2)
63     self.conv2_2 = ConvBlock(nb_filter[2]*2 + nb_filter[3],
64                               nb_filter[2])
65
66     self.ag0_3 = AttentionGate(nb_filter[1], nb_filter[0],
67                                nb_filter[0]//2)
68     self.conv0_3 = ConvBlock(nb_filter[0]*3 + nb_filter[1],
69                               nb_filter[0])
70     self.ag1_3 = AttentionGate(nb_filter[2], nb_filter[1],
71                                nb_filter[1]//2)
72     self.conv1_3 = ConvBlock(nb_filter[1]*3 + nb_filter[2],
73                               nb_filter[1])
74
75     self.ag0_4 = AttentionGate(nb_filter[1], nb_filter[0],
76                                nb_filter[0]//2)
77     self.conv0_4 = ConvBlock(nb_filter[0]*4 + nb_filter[1],
78                               nb_filter[0])
79
80     self.final1 = nn.Conv2d(nb_filter[0], num_classes, kernel_size
81                               =1)
82     self.final2 = nn.Conv2d(nb_filter[0], num_classes, kernel_size
83                               =1)
84     self.final3 = nn.Conv2d(nb_filter[0], num_classes, kernel_size
85                               =1)
86     self.final4 = nn.Conv2d(nb_filter[0], num_classes, kernel_size
87                               =1)
88
89     def forward(self, input):
90         x0_0 = self.conv0_0(input)
91         x1_0 = self.conv1_0(self.pool(x0_0))
92         x2_0 = self.conv2_0(self.pool(x1_0))
93         x3_0 = self.conv3_0(self.pool(x2_0))

```

```

78     x4_0 = self.conv4_0(self.pool(x3_0))
79
80     g0_1 = self.up(x1_0)
81     x0_1 = self.conv0_1(torch.cat([self.ag0_1(g0_1, x0_0), g0_1],
82                                   1))
83     g1_1 = self.up(x2_0)
84     x1_1 = self.conv1_1(torch.cat([self.ag1_1(g1_1, x1_0), g1_1],
85                                   1))
86     g2_1 = self.up(x3_0)
87     x2_1 = self.conv2_1(torch.cat([self.ag2_1(g2_1, x2_0), g2_1],
88                                   1))
89     g3_1 = self.up(x4_0)
90     x3_1 = self.conv3_1(torch.cat([self.ag3_1(g3_1, x3_0), g3_1],
91                                   1))
92
93     g0_2 = self.up(x1_1)
94     x0_2 = self.conv0_2(torch.cat([x0_0, self.ag0_2(g0_2, x0_1),
95                                   g0_2], 1))
96     g1_2 = self.up(x2_1)
97     x1_2 = self.conv1_2(torch.cat([x1_0, self.ag1_2(g1_2, x1_1),
98                                   g1_2], 1))
99     g2_2 = self.up(x3_1)
100    x2_2 = self.conv2_2(torch.cat([x2_0, self.ag2_2(g2_2, x2_1),
101                                   g2_2], 1))
102
103    g0_3 = self.up(x1_2)
104    x0_3 = self.conv0_3(torch.cat([x0_0, x0_1, self.ag0_3(g0_3,
105                                   x0_2), g0_3], 1))
106    g1_3 = self.up(x2_2)
107    x1_3 = self.conv1_3(torch.cat([x1_0, x1_1, self.ag1_3(g1_3,
108                                   x1_2), g1_3], 1))
109
110    g0_4 = self.up(x1_3)
111    x0_4 = self.conv0_4(torch.cat([x0_0, x0_1, x0_2, self.ag0_4(
112                                   g0_4, x0_3), g0_4], 1))
113
114    output1 = self.final1(x0_1)
115    output2 = self.final2(x0_2)
116    output3 = self.final3(x0_3)
117    output4 = self.final4(x0_4)
118

```

```

109         if self.deep_supervision:
110             return [output1, output2, output3, output4]
111         else:
112             return output4

```

Listing 8.1: Attention U-Net++ Implementation (PyTorch)

A.3 Source Code – Training Loop

The following snippet shows how the model is instantiated and trained using deep supervision.

```

1 model = Attention_UNet_PlusPlus(in_channels=1, num_classes=1,
    deep_supervision=True).to(device)
2 criterion = DeepSupervisionLoss()
3 optimizer = optim.AdamW(model.parameters(), lr=LEARNING_RATE,
    weight_decay=1e-4)
4 scheduler = CosineAnnealingLR(optimizer, T_max=EPOCHS, eta_min=1e-6)
5
6 best_val_dice = 0.0
7
8 for epoch in range(EPOCHS):
9     print(f"\n--- Epoch_{epoch+1}/{EPOCHS} ---")
10
11     # Train
12     model.train()
13     train_loss = 0.0
14     for images, masks in tqdm(train_loader, desc="Training"):
15         images, masks = images.to(device), masks.to(device)
16
17         optimizer.zero_grad()
18         outputs = model(images)
19         loss = criterion(outputs, masks)
20
21         loss.backward()
22         optimizer.step()
23         train_loss += loss.item()
24
25     scheduler.step()
26
27     # Validate

```

```

28 model.eval()
29 val_loss = 0.0
30 val_dice = 0.0
31 with torch.no_grad():
32     for images, masks in tqdm(val_loader, desc="Validating"):
33         images, masks = images.to(device), masks.to(device)
34         outputs = model(images)
35
36         final_pred = outputs[-1]
37         val_loss += DiceBCELoss()(final_pred, masks).item()
38         val_dice += calculate_dice(final_pred, masks)
39
40 avg_train_loss = train_loss / len(train_loader)
41 avg_val_loss = val_loss / len(val_loader)
42 avg_val_dice = val_dice / len(val_loader)
43
44 print(f"Train_Loss:_{avg_train_loss:.4f}|_Val_Loss:_{avg_val_loss:.4f}|_Val_Dice:_{avg_val_dice:.4f}")
45
46 if avg_val_dice > best_val_dice:
47     best_val_dice = avg_val_dice
48     torch.save(model.state_dict(), "attention_unet_plusplus_kits19.pth")
49     print(f"_Saved_new_best_model_with_Dice:_{best_val_dice:.4f}!")

```

Listing 8.2: Training Script for Attention U-Net++

VIII_A_DRL

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Figure 8.1: Similarity Report Overview

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10	Shylesha V. Channapattana, Srinidhi Campli, A. Madhusudhan, Srihari Notla, Rachayya Arkerimath, Mukesh Kumar Tripathi. "Energy analysis of DI-CI engine with nickel oxide nanoparticle added azadirachta indica biofuel at different static injection timing based on exergy", Energy, 2023 Publication	1 %
11	Yaqin Zhou, Chang Xu, Yunfeng Dai, Xingming Feng, Yunpeng Ma, Qingwu Li. "Dual-View Stereovision-Guided Automatic Inspection System for Overhead Transmission Line Corridor", Remote Sensing, 2022 Publication	<1 %
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Figure 8.2: Similarity Report Overview

9. PUBLICATION PAPER DRAFT

Overview

As part of this project, a research paper has been prepared based on the proposed Attention U-Net++ with Deep Supervision model for kidney CT segmentation. The paper presents a comprehensive overview of the methodology, including the architecture design, attention mechanisms, and deep supervision strategy employed to enhance segmentation performance. It also discusses the experimental setup, dataset details, evaluation metrics, and comparative analysis with existing state-of-the-art models.

The primary objective of the paper is to highlight the effectiveness of the proposed model in addressing key challenges in medical image segmentation, such as low contrast, noise, and anatomical variability. The results demonstrate significant improvements in segmentation accuracy and robustness, making the work suitable for submission to a peer-reviewed journal or conference in the field of medical image analysis.

Paper Details

- **Title:** Attention U-Net++ with Deep Supervision for Robust Kidney Segmentation in CT Images
- **Authors:** V. Vignesh, T. Lasya Priya, DRL Prasanna
- **Affiliation:** Department of Information Technology, Vasavi College of Engineering, Hyderabad, India
- **Dataset:** CHAOS CT Dataset
- **Key Result:** Dice Similarity Coefficient (DSC) = 0.981, outperforming U-Net, Attention U-Net, and U-Net++
- **Contribution:** Integration of attention mechanisms with U-Net++ and deep supervision to improve segmentation accuracy and robustness
- **Application:** Kidney segmentation for medical diagnosis, treatment planning, and disease monitoring

Paper Draft The following figure shows the first page of the prepared publication paper draft:

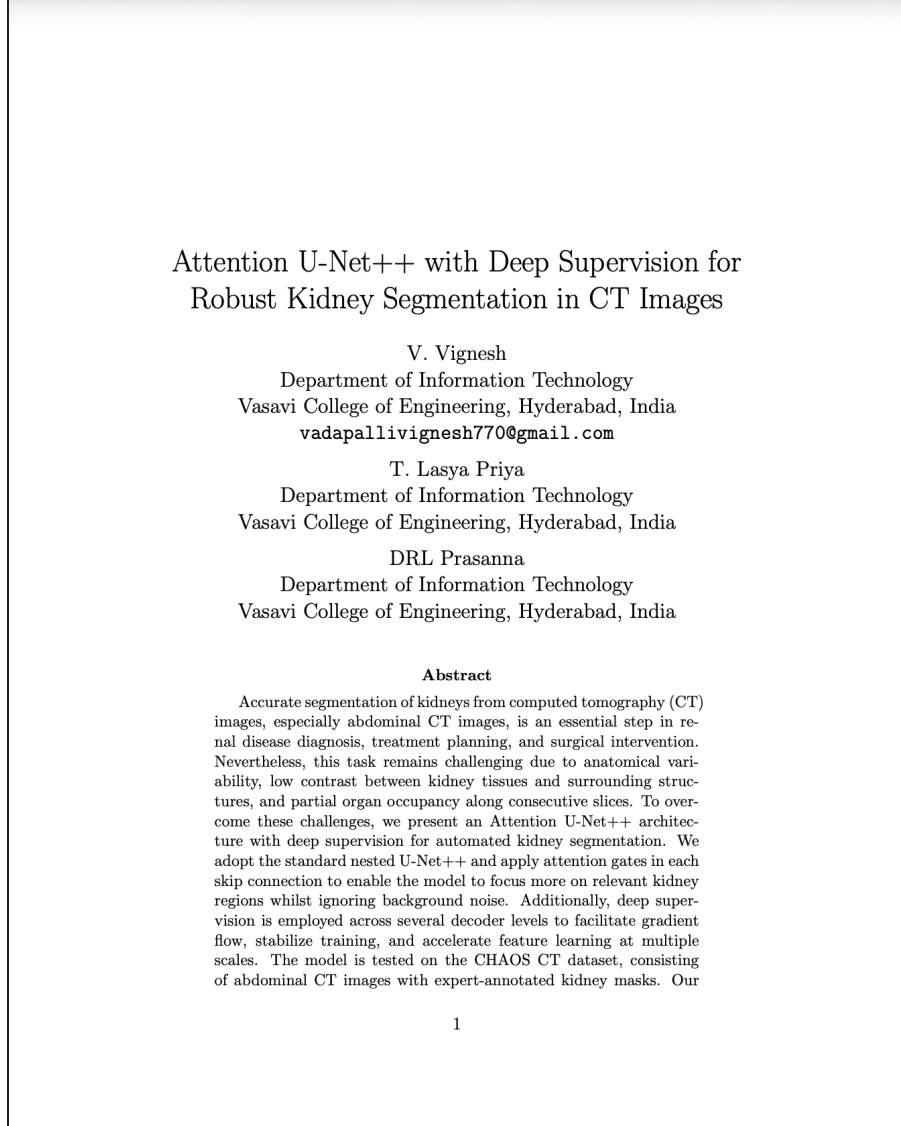


Figure 9.1: First Page of the Publication Paper Draft – Attention U-Net++ for Kidney CT Segmentation

The paper presents an Attention U-Net++ architecture with deep supervision for automated kidney segmentation from abdominal CT images. The model is evaluated on the CHAOS CT dataset and achieves a Dice Similarity Coefficient (DSC) of **0.981**, outperforming baseline architectures including standard U-Net (0.88), Attention U-Net (0.91), and U-Net++ (0.92). The paper details the nested U-Net++ structure, the attention gate mechanism, and the role of multi-scale deep supervision in stabilizing training and improving feature learning.

10. PROGRAM OUTCOMES (PO)

PO Mapping Program Outcomes (POs) are the knowledge, skills, and attitudes students are expected to attain by graduation. The table below maps project outcomes to PO1–PO11.

Table 10.1: PO Mapping Table

Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
1. Design and development of Attention U-Net++ architecture for kidney segmentation	3		3		3						2
2. Experimental validation using patient-wise split, augmentation and cross-validation		3		3	3						2
3. Application of AI for clinical decision support and healthcare improvement			2			3					

Scale: 3 – Strong, 2 – Moderate, 1 – Weak, Blank – No Mapping

PO Justification

- **PO1 (Engineering Knowledge):** Deep knowledge of CNNs and medical image processing is applied in designing Attention U-Net++.
- **PO2 (Problem Analysis):** Challenges like low contrast and data imbalance are systematically analyzed and addressed.
- **PO3 (Design/Development):** The proposed architecture is a designed engineering solution for automated kidney segmentation.
- **PO4 (Conduct Investigations):** Patient-wise split and cross-validation are systematic investigation techniques.
- **PO5 (Modern Tool Usage):** TensorFlow, Keras, Python, NumPy, and Matplotlib are used throughout.

11. PROGRAM SPECIFIC OUTCOMES (PSO)

PSO Mapping Program Specific Outcomes (PSOs) define specialized skills expected from Information Technology graduates. The table below maps project outcomes to PSO1–PSO3.

Table 11.1: PSO Mapping Table

Outcomes	PSO1	PSO2	PSO3
1. Development of AI-based medical imaging software system	3		2
2. Data preprocessing, augmentation and performance optimization techniques	3		2
3. Research-driven model enhancement and comparative evaluation	2	2	3

Scale: 3 – Strong, 2 – Moderate, 1 – Weak, Blank – No Mapping

PSO Justification

- **PSO1 – AI/ML-based Software Development:** Development of the Attention U-Net++ system directly aligns with PSO1, which emphasizes designing and developing intelligent software systems using AI/ML.
- **PSO2 – Data and System Management:** Implementation and management of medical imaging datasets with preprocessing and validation demonstrates competency in handling real-world data pipelines.
- **PSO3 – Research and Evaluation:** Comparative analysis and experimental evaluation of model architectures reflects PSO3, focusing on research-driven problem solving.

12. SUSTAINABLE DEVELOPMENT GOALS (SDG)

SDG Mapping The project contributes to the United Nations Sustainable Development Goals as follows:

Table 12.1: SDG Goals Mapping Table

Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Deep Learning Approach for Kidney CT Segmentation Using Attention U-Net++			3						2								

Scale: 3 – Strong, 2 – Moderate, 1 – Weak, Blank – No Contribution

SDG Justification

SDG 3 – Good Health and Well-Being (Level: 3)

- The AI-based kidney segmentation system supports accurate medical diagnosis and early detection of renal diseases, directly improving healthcare outcomes.
- Automated segmentation reduces the diagnostic workload on radiologists, minimizing errors and improving patient care quality.
- The system contributes to accessible, reliable, and consistent healthcare analysis tools aligned with SDG 3.

SDG 9 – Industry, Innovation and Infrastructure (Level: 2)

- Development of the Attention U-Net++ architecture promotes innovation in AI-driven medical imaging technology and intelligent healthcare infrastructure.